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SIGNIFICANT SMALL CRAFT OF 1999

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Front cover: Patrol vessel, Roebuck Bay; seismic research ship Geco Eagle; pilot craft Humber Callisto.





AFRIKA: pelagic factory trawler heads for the tropics

Afrika, one of the most recent pelagic stern trawlers to join the Dutch fleet, is designed to operate in the tropics, fishing mainly for sardinella and horse mackerel in West African waters and operating for much of the year from the Canary Islands.

Built by YVC at the Capelle-aan-den-IJssel yard, the 6400dwt vessel has an overall length of 126.20m and a beam of 11.02m. Built on much the same lines as conventional freezer trawlers in the Dutch fleet, it has a long foredeck with superstructure close to the stern, leaving a trawl deck aft of the wheelhouse. Accommodation, mess and galley spaces are located in the superstructure, with the twin level factory deck in the aft section.

An MAK 6M601C main engine, developing 7,500kW at 425 rev/min turns a Berg 3,800mm diameter CP propeller at 166 rev/min inside a Hodi nozzle via a Renk Tacke gearbox. This configuration gave a speed of 18 knots on sea trials. A 700kW bow thruster is also fitted.

Auxiliary power is also derived from MAK units - a 9M25 set, rated 2,610kW at 720 rev/min and an 8M20, rated 1,360 kW at 900 rev/min. An AC Leroy Somer shaft generator develops 4,200kVA and a DC Bakker Slidrecht shaft generator produces 770kW.

Over 1,400 tonnes of heavy fuel and 280 tonnes of gas oil can be carried, while ballast tanks will hold 805 tonnes and the

fresh water tanks on board have a 120-tonne capacity.

The fish room, which is on two levels, with a forklift on board for managing frozen products, can hold 172,100 cartons of sea frozen fish. A pair of Hydraskin cranes is mounted on the port side of the foredeck for unloading catches through hatches that are offset to port, minimising disruption to the factory deck on the level below.

Fishing gear is handled over the stern using winches supplied by Brusselle Marine Industries. The twin trawl winches, powered by Bakker Slidrecht 350kW motors, each have a 4,000m capacity and an eleventh layer pull of 510kN. Twin net drums are used to hold two of the vessel's five trawls ready for use. Other gear is stored aft, with two more Hydraskin cranes located on the port side of the trawl deck and at the top of the stern gantry for handling fishing gear.

Additional winches include a pair of cod end winches from Brusselle and two net sounder cable winches from Karmøy mounted just aft of the wheelhouse. Deck operations are commanded from a console at the back of the wheelhouse with a full view over the deck.

All of the winches are controlled from this position, and a PA system is fitted for communication with the deck crew. A Marelec trawl control system is fitted to monitor warp tension and length, and this

has a monitor in the central wheelhouse console, with a repeater at the deck control position.

Catches taken by vessels of this type are relatively sensitive species such as sardinella and mackerel, and instead of bringing bags of fish on board, catches are pumped aboard using a Karmøy fish pump. The fish are taken through a dewatering system before passing to the tanks below, which have a maximum holding capacity of 540 tonnes.

The Grenco refrigeration system on board makes it possible to chill the maximum capacity through 32°C in eight hours. A slurry ice plant is capable of producing 40 tonnes/day, and this is used for cooling fish in the tanks and the intermediate cooling bins. The whole of the refrigeration system in the ship can be monitored and controlled from the engineer's control room.

The twin level factory deck was supplied by Machinefabriek van Rijn, and has a vacuum pump system for transporting fish. Fish are brought from the tanks via a grading section where they can be sorted according to species and weight and held in the intermediate cooling bins.

All of the catch is frozen in Jackstone vertical plate freezers served by three lanes of overhead conveyors. A Marelec quality control station is fitted next to the grader, allowing the factory deck crew to keep

constant track of the grading weights. There are 24 vertical plate freezers on the lower factory deck, and a further 16 freezers on the deck above. Fish are carried to the upper deck using an extended series of conveyors, and are delivered to the freezers in the same way.

Frozen blocks from the freezers are carried to a packing area along a secondary system of conveyors at waist level, where they are boxed and strapped. An extended conveyor brings frozen blocks down from the top factory deck. Each block is weighed individually, and a computer-controlled inkjet is used to mark each box with its contents. The boxing machine has been built to run with a minimum of manpower, and only needs to be loaded with empty boxes to run automatically. It is controlled using a PLC (Programmable Logic Control) computer, and a similar control unit is used to manage the factory deck's vacuum pumps.

An AFAK Techniek wrapping machine has been fitted, and this seals each carton of fish into a plastic wrapper. When strapped and marked, the boxes are carried by lift down to the fish rooms and stowed on pallets. Depending on the type of fish, and the mix of species, a crew of ten should be able to run the

factory deck at full capacity. Freezing time for each freezer is four hours, giving a total freezing capacity of 350 tonnes per day.

The wheelhouse is laid out with a control console aft for the deck machinery, and a horseshoe console forward. Side by side skippers' chairs have a full view of the twin Furuno radar screens, and an array of fish finding equipment. Two colour sounders are from JRC, and a Furuno paper sounder have their displays built into the console. A Furuno colour scanning sonar is at the centre, and there are further displays for the trawl mounted Wesmar and Simrad sonar sets. A Scanmar cableless system is also fitted for monitoring catches. Controls for the sonars are built into the central console between the two skippers' seats, along with throttle and steering controls.

A pair of chartplotters linked to DGPS positioning systems is provided with displays in the control console, and another GPS receiver and a navtex receiver are located by the chart table

VHF and MF/HF communications sets are built into the port side of the console, and a separate communications room on the port side of the wheelhouse contains the satellite communications equipment, as well as fax and weather fax.

Main Particulars

Owner

Jacson BV, The Netherlands

Bullder

YVC, The Netherlands

Length oa 126.20m

Beam oa 11.02m

Main engine

MAK 6M601C diesel, developing 7,500kW

Propulsion

Berg CP propeller inside a Hodi nozzle

Main generators

MAK 9M25 set, rated 2,610kW & MAK

8M20 set, rated 1,360kW

Tank capacities

Heavy fuel 1,400t

Gas oil 280t

Ballast 805t

Potable water 120t

Service speed 18 knots

Deadweight 6,400t

Equipment

4 x Hydramaskin cranes, Brusselle Marine winches, Karmoy cable winches and fish pump, Marelec trawl control system, Gresco refrigeration system, AFAK Techniek wrapping machine, Furuno radar and sounder, JRC colour sounders, Wesmar and Simrad sonar sets

River Runner

150 SERIES / 200 SERIES

RR75	RR150
RR200	RR300

- Low wash
- Low noise
- Low emmisions
- Low fuel consumption





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ATLANTIDE: 1930's yacht sails again

The 34.8m sailing yacht *Atlantide* deserves merit amongst significant small vessel deliveries for 1999, not because it is innovative in design but because of the painstaking effort involved in restoring it. When owner, Thomas Perkins, a venture capitalist from California discovered the vessel in Malta it was in a deplorable condition. Some two years on, however, it floats again with all of the elegance and charm it flaunted in the 1930s.

Built to an original Alfred Mylne design, *Atlantide* has had a colourful history since being launched at the Phillips & Son yard in Dartmouth, including participation in the Dunkirk evacuation as part of the renowned Little Ships Armada.

With naval architect Ken Freivokh tasked to produce restoration plans, Manoel Island Yacht Yard (the small boat wing of Malta Drydocks) was asked to repair the steelwork. In the event, some 90 per cent of the steel in the hull had to be replaced, a task which took a year to complete.

The interior of the yacht was conceived by the Freivokh team in 1930's art deco style to match the period of its early days. This involved use of elaborate marquetry and Lalique glass and the considerable research carried out inspired some unique artwork, as can be seen in the internal panelling. A notable example is the ornate nymph sand-blasted into glass on one of the dining room bulkheads. This is

Main Particulars

Owner

Thomas Perkins, USA

Restoration

Manoel Island Yacht Yard, Malta,
Camper & Nicholson's, UK &
C&C Designs, UK

Designer (restoration)

Ken Freivokh, UK

Length oa 34.8m

Beam oa 6.1m

Main engines

2 x Gardner 8L3B vintage units

Crew 8

Equipment

Anschutz autopilot and gyrocompass,
Furuno radar, Leica GPS,
Servowatch monitoring system



surrounded by Lalique light ramps, inlay marquetry in Madrona and Macassar ebony and fibre optics. Every part of the yacht has been given the same attention to detail, including the use of abalone and mother of pearl inlays.



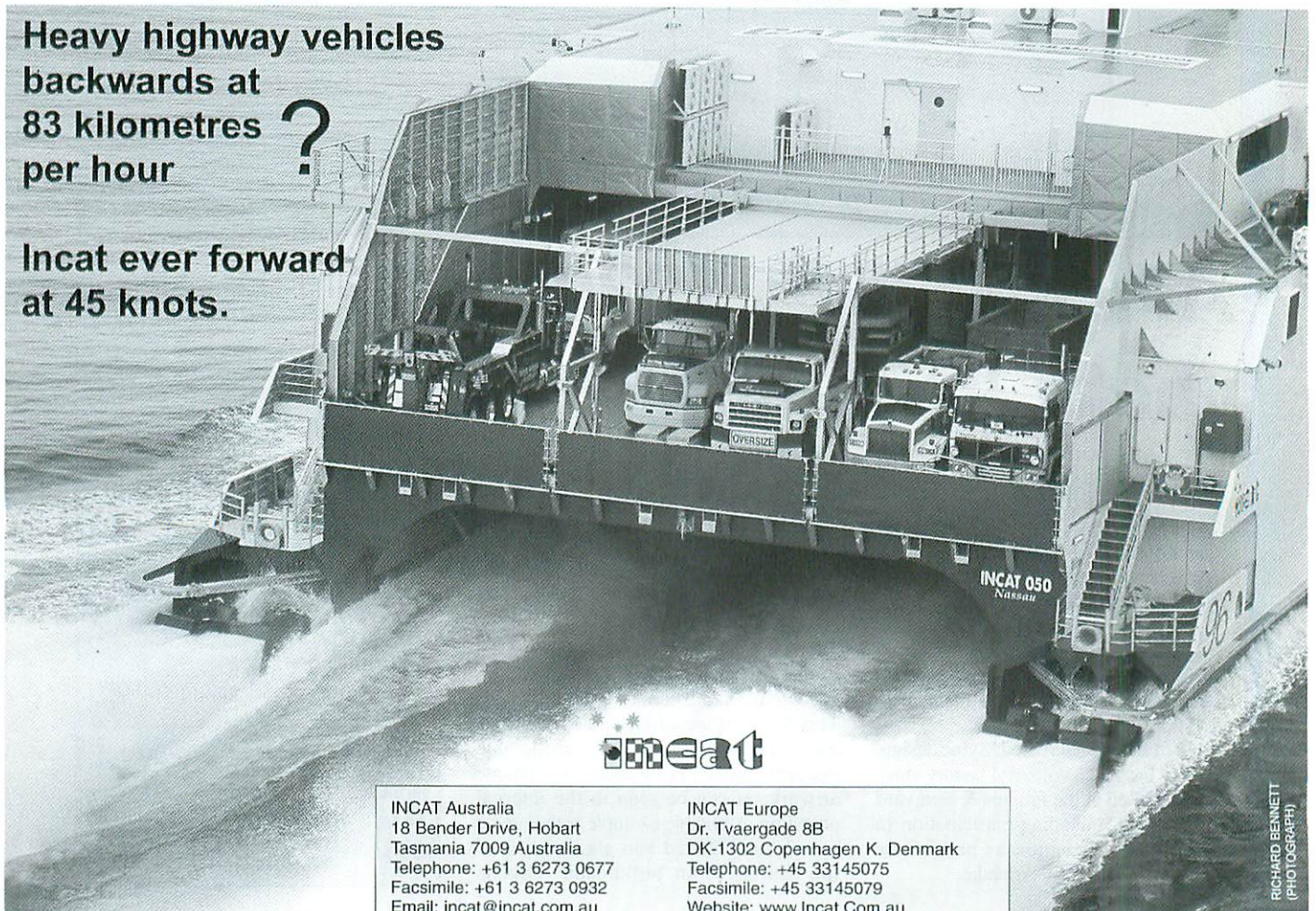
In 1998 *Atlantide* was shipped as deck cargo to the UK yard of Camper & Nicholson's. Here, craftsmen could implement the ambitious designs and engineers could work on the onboard machinery. This included removing and refitting the two 230hp vintage Gardner 8L3B engines, deck work, systems installation, creation of a new teak wheelhouse and major work to form a fully fitted galley and extensive crew quarters for eight.

C&C Designs, based in Norfolk, UK, was appointed to fit out the social areas of the vessel based on its previous success in refitting Paul Getty's yacht *Talitha G*. The project involved the installation of 1,500ft³ of teak, 14 miles of cable and 470 lights. This element of the project also took 12 months.

Now fully restored and embellished, *Atlantide* will be based in Antibes and has a job of work to do - acting as tender to the owner's other pride and joy, a 1915 classic racing schooner *Mariette*. To enable it to perform its duties, the wheelhouse is deceptively well-equipped with an Anschutz autopilot and gyrocompass, Furuno radar and Leica GPS. In modern style, all the machinery can be monitored from the wheelhouse through a Servovatch system.

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backwards at
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per hour** ?

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at 45 knots.**



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RICHARD BENNETT
(PHOTOGRAPH)



BOHENGY: Solent stretch fulfils Bahamas brief

The Solent Class catamaran, a well-proven all-aluminium design from UK yard FBM Marine, has undergone a 2m length extension to increase its passenger carrying capacity. The Mark 2 version, first put into service in Hong Kong on a commuter route to Shenzhen, has also joined a fleet in the Caribbean, having been further modified by FBM licensed builder PRS - Pequot River Shipworks in the USA.

With uniquely colourful livery, *Bohengy* has been equipped to fulfil local community needs such as cargo transfer in the Bahamas as well as providing service as a tourist ferry. Outside seating on the aft end of the upper deck abaft the wheelhouse has been installed (a first for a Solent Class vessel) and a large refrigerated storeroom incorporated at the port aft end of the main cabin. Primarily designed for the inter-island transport of perishable foodstuff, this room has a large horizontal exterior door sized to take an 8ft coffin. A large baggage store is also provided for the transport of white goods and hanging rails for formal clothing.

Although fitted with a total of 195 seats, maximum passenger capacity is 178 in keeping with the Passenger Safety Certificate of the Bahamas Marine Authority. The main lounge provides fully enclosed saloon style accommodation for 144 mostly in 3-4-3 and 3-5-3 aircraft style arrangements. The remainder of the seating is located on the wheelhouse deck with 33 seats fully protected by a rear extension to

the wheelhouse roof and clear PVC side screens. An extra 18 exterior seats can be used by those passengers preferring to sit outside. A servery, located amidships between the port and starboard boarding points, is available for snacks, soft drinks and duty-free alcoholic beverages, which are sold for onboard consumption only. The area aft of the servery has tables and seating in a café style arrangement.

Power is provided by a pair of Detroit Diesel 12V 4000 engines, each rated 2,330hp at 1,900 rev/min. These drive Kamewa KMW 63 waterjets via ZF gears to give an estimated full load speed of 33 knots at 100 per cent MCR. Each engine is sea-water cooled, flexibly-mounted, electrically started and fitted with a 24 volt alternator. An integrated engine/jet KMW control system is fitted utilising twin KMW combinator levers, one set each at the captain's and engineer's helm position in the wheelhouse. Back-up control is incorporated at the captain's position and local emergency control in the engine room.

Bohengy is fitted with two AC generators, continuously rated at approximately 52kW, with four-cylinder Perkins T4236 water-cooled flexibly mounted prime movers. These run at 1,800 rev/min and are capable of 10 per cent overload for one hour in any 12-hour period. The diesel generators can be electrically started locally and at the main switchboard and monitored from the

wheelhouse with engine-mounted local control. Each machine is able to supply the normal winter seagoing load (excluding air conditioning) but units can be run in parallel for change-over purposes and in the summer when power for air conditioning is required.

Main Particulars

Builder

Pequot River Shipworks, USA

Designer

FBM Marine, UK

Length oa 35.00m

Length wl 29.50m

Beam 8.32m

Draft 1.14m

Main engines

2 x Detroit Diesel 12V4000,
1,738kW each

Propulsion

2 x Kamewa KMW 63 waterjets

Auxiliary Power

Perkins T4236 diesel generators,
1,800 rev/min each

Speed 34 knots

Range 160nm

Passengers 178

Classification

DNV +1A1, HSLC, EO, R3. IMO
HSC Code



BONANZA EXPRESS: **Weight-saving techniques** **make for speedy Ro-Pax**

Built for operation in the Canary Islands by leading Spanish ferry operator Lineas Fred Olsen SA, *Bonanza Express* was launched at the end of January '99 from Incat Tasmania's giant Coverdales shipbuilding facility at Prince of Wales Bay near Hobart.

The 96m wavepiercing passenger/vehicle catamaran is a near sister of Incat's first dedicated Ro-Pax vessel *DevilCat*, which now operates on TT Line's summer service between Melbourne and George Town, Australia. *Bonanza Express* entered service last April on the 35nm route between the ports of Santa Cruz de Tenerife and Agaete on the nearby island of Gran Canaria, reducing travel time between the two islands from two hours 15 minutes to just 60 minutes.

Flexibility and payload are the characteristics that most distinguish Incat's first 96m vessels and *Bonanza Express* incorporates several new features enabling it to transport significant numbers of heavy vehicles.

Perhaps the most remarkable fact about the ship is that it is almost capable of carrying its own weight. This equates to

around 800dwt, which can be carried as a broad mix of light and heavy road freight. Such an achievement has been made possible by a diligent approach to weight minimisation during construction, coupled with a subtle re-design of the 91m class hull.

Weight reduction is largely due to the extensive use of aluminium extrusion. An extruded hollow box planking was developed to ensure nine-tonne axle loadings for the main vehicle deck and tunnel underside plating is strong yet light in weight due to the use of corrugated aluminium. Other savings were made with non-structural superstructure side panels, which were bonded with adhesives rather than by welding and could, therefore, be thinner and lighter.

Designated as a Car Ferry B Incat 051, the vessel has open foredeck and stern, allowing adequate natural ventilation to the vehicle decks. This policy led to a considerable saving, not only in terms of structural weight but also by permitting the omission of normally mandatory items such as structural fire protection, ventilation fans, fire dampers, overhead

fixed sprinkler systems and additional lighting.

Further structural weight savings were achieved by reducing the physical size of the passenger cabin and wheelhouse. With a total capacity of 755 persons, most of the passenger accommodation is on a single tier with the wheelhouse positioned well aft, above the exclusive panoramic sky lounge.

To cater for the increase in passenger numbers over the first 96m vessel, *Bonanza Express* has an enlarged sky lounge incorporating a café-bar, crew facilities and toilets. Seating is arranged in pairs with the forward part of the area, designated for first class passengers, comprising 62 Beurteaux Ocean Executive aircraft style seats. Double seating also extends aft along each side of the vessel below the wheelhouse.

A central stairway forward leads down to Tier 2, emerging just aft of the self-service bar and café where travellers can buy beer and spirits as well as a wide variety of hot and cold food. Accommodation comprises Beurteaux tub style seating in clusters of four around

wood-grain laminate tables. The main seating area features cushioned recliners in 2-4-2 configurations and the forward lounge, the most popular seating area due to its views over the ship's bow, has 158 seats in pairs.

Vehicle access is via a shore-based stern ramp or ramps, the transom arrangements being designed to allow ramps providing a minimum of four lanes. In total they offer 330 lane metres at 2.7m wide x 4.3m high for heavy, high and wide vehicles and 370 car lane metres at 2.3m wide. This gives *Bonanza Express* the flexibility to carry 240 cars and no heavy vehicles, 12 road freight trailers with 180 cars or 24 road trailers with 85 cars. While the maximum deadweight capacity of 800 tonnes is one key to achieving these freight loads, it is the provision of moveable mezzanine vehicle decks that allows the ship to offer the necessary lane metres required for maximum car loading as well as offering the headroom demanded by oversize freight vehicles.

Deck officers operate the vessel more like an aircraft than a traditional ocean-going ship. A surprisingly small raised control station has large outward raked windows fore and aft to reduce glare and give all-round visibility. A central aft-facing docking console in

the control station obviates the need for bridge wings and their associated structural weight.

The station is fitted with the latest in electronic, navigation and communication equipment to comply with High Speed Craft Code Sea Area A2. During normal operations the captain, navigator and engineer sit abreast at the main console where the former two can monitor a dedicated inter-switched Litton high speed radar interfaced to the Anschutz gyro compass, C-Plath magnetic compass and one of two Litton LMX 400 GPS units. The engineer is positioned behind the dual screens of the latest ISIS 2500 integrated alarm and assessment system from Litton Marine Systems.

As with previous Incat wavepiercers, *Bonanza Express* is diesel powered, in this instance by four Alstom-Ruston 20RK270s developing 7,080kW each at 1,030 rev/min. These drive transom-mounted steerable Lips 150 D waterjets via Reintjes VLJ6831 gearboxes. The engine room in each hull houses two of the main propulsion diesels and two auxiliaries, which are mounted transversely in the forepart of the space. All four waterjets are configured for steering and reverse functions.

Main Particulars

Operator

Lineas Fred Olsen, Canary Islands

Builder

Incat Tasmania Pty, Australia

Length oa 96.00m

Length wl 86.00m

Beam oa (ex Fenders) 26.00m

Draft 3.70m

Main engines

4 x Alstom/Ruston 20RK270 diesels,

7,080kW each

Propulsion

4 x Lips 150 D waterjets configured for steering and reverse

Passengers & crew 755

Vehicle deck capacity 330m truck lane -

2.7m wide x 4.3m high 425m car lane -

2.3m wide

Speed (Lightship) 48 knots

Speed (Loaded/800t) 37.5 knots

Classification

DNV +1A1 HSLC R1 Car Ferry "B" EO Cert

Equipment

Beurteaux seating, Litton radars, GPS and ISIS

2500 systems, Anschutz gyro compass, C-Plath

magnetic compass, Lipstronic jet control, Liferaft

Systems Australia marine evacuation system,

Colbeck & Gunton fire protection



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DC 1500: Hong Kong five give chase to smugglers

Designing a sea-kindly, low noise fast interceptor craft for the Hong Kong Marine Police called for the extensive boatbuilding skills of Dutch yard Damen Shipyards and the offshore power boat expertise of UK firm Cougartek Design. The result, designated the Damen Cougartek 1500, is capable of speeds in excess of 60 knots, making it well able to catch up with smugglers. Five vessels have been completed.

Built in accordance with Damen's ISO 9001 quality regulations with supervision from surveyors of the Hong Kong Marine Department, the ultra fast DC 1500s measure 14.57m x 2.91m x 1.80m. An aluminium superstructure was selected in preference to materials such as FRP for durability and ease of maintenance and retrofit. A deepV hard chine hull form has been incorporated to enhance sea-keeping and handling and to provide a safe working environment when the craft is racing at high speeds.

Power is supplied by three Mercruiser HP 500 Bulldog gasoline engines driving four Bravo One HP sterndrives via cardan

shafts. A combined output of 1,119kW is achieved. Noise reduction at speeds of up to 45 knots is aided by a special exhaust system which forces the exhausts to discharge under water, through the sterndrives. When higher speeds are required, the exhausts switch to discharging through the transom.

Due to the demanding role of the vessels, maintenance of the engines is made simple and efficiency is aided by a quick replacement system, whereby a new unit can be installed within the hour using a vertical lift-in/out procedure. Fittings used include special connectors, which can be simply plugged and unplugged. All three engines are identical, thus minimising the level of stock and spare parts required. This feature also extends to other essential equipment such as the electrical bilge pumps, which are also fitted with plugs for ease of installation/replacement. The vessels are equipped with a set of lifting eyes, which when combined with a sling, allow a single-point lift-out onto a custom-designed wheeled storage cradle. When in place, personnel have easy access to the

Main Particulars

Owner

Hong Kong Marine Police

Builder

Damen Shipyards, The Netherlands

Designer

Cougartek Design, UK & Damen Shipyards, The Netherlands

Length oa 14.57m

Length wl 12.01m

Beam oa 2.91m

Beam mld 2.71m

Draft 1.80m

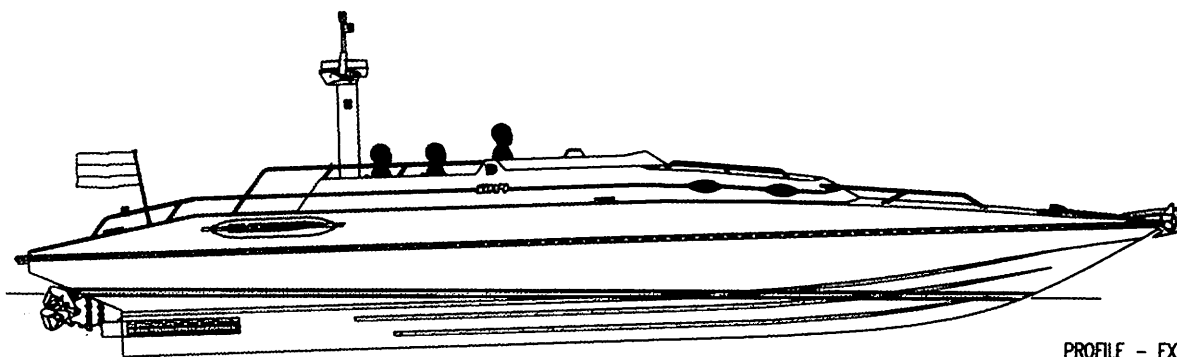
Propulsion

3 x Mercruiser HP 500 Bulldog gasoline engines driving 4 x Bravo One HP sterndrives via cardan shafts. Combined output of 1,119kW

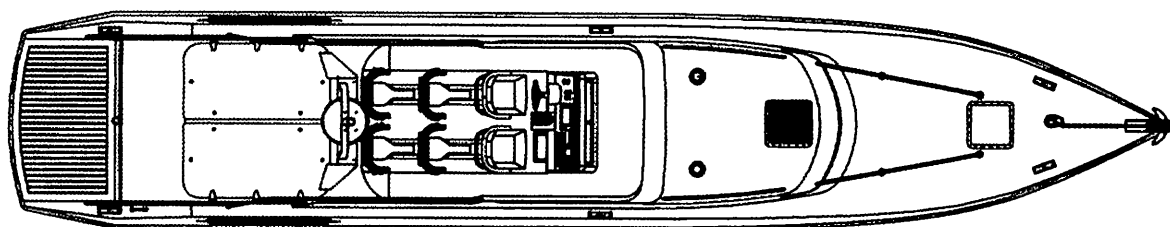
Crew (maximum) 8

Cruising speed 42 knots

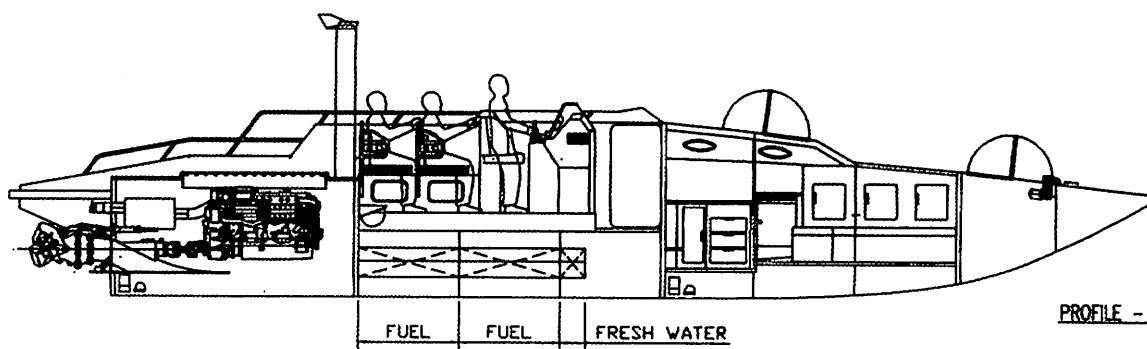
Interdiction speed . 60 knots plus



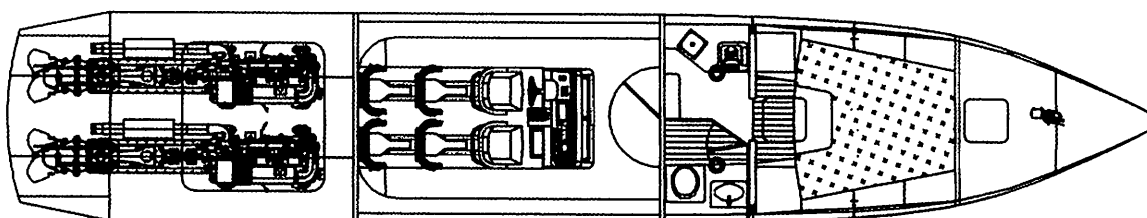
PROFILE - EXTERNAL



PLAN VIEW - EXTERNAL



PROFILE - INTERNAL



PLAN VIEW - INTERNAL

stern drives, propellers and steering system.

A combination of bolster and saddle type seating is fitted to accommodate a crew of up to eight although five is the norm. All seats are fitted with backrests for secure positioning and are located well inboard in the self-draining cockpit. Control consoles are located aft and forward and are surrounded by a walkway for unrestricted movement, essential when preparing to board vessels.

The forward console has seating for two, with helmsman positioned to port and boarding officer to starboard. Gearshift and throttle controls are fitted centrally and the arrangement provides unobstructed views over the bow. The

aft console is also for two and houses the monitoring equipment for the propulsion system, as well as an extensive array of nav aids. Communication equipment is once again fitted centrally and general communication between crew is via helmet-mounted voicelucers.

Damen has also introduced a great deal of flexibility in the onboard set-up, with the option for a number of differing cabin layouts to suit particular customer operating requirements e.g. choice between open cockpit and fully-enclosed cabin. Such flexibility, coupled with the safe working environment and precise handling - the craft is capable of maintaining a speed in excess of 45 knots

in Sea State 3 conditions - enables a lot of work to be undertaken at close quarters.

The vessels will usually operate in pairs with one utilised as the lead interdiction boat, while the other provides crucial back-up. As well as operating on a stand-alone basis, the DC 1500 can also be used in a mother and daughter arrangement, linked up, for example, with a patrol vessel acting in a surveillance, command/control role.

The last two boats in the series were commissioned in October '99 and the completion of the contract continues the strong relationship between Damen and China, with the Dutch yard having already built more than 30 patrol vessels for the Hong Kong police and customs authorities.



***FAR SOVEREIGN*: packed with power for offshore tasks**

Major offshore vessel operator, Farstad Shipping, added the powerful UT 741 AHTS/towing vessel *Far Sovereign* to its fleet in late '99. Multi-functional in order to undertake a wide range of offshore support tasks, the vessel is equipped with a main towing winch capable of 500 tonnes maximum pull and has a bollard pull of 280 tonnes.

Measuring 85.20m x 20.50m, *Far Sovereign* was built in Norway at the Langsten Slip yard in Tomrefjord and has been operated in recent months by European Marine Contractors (EMC) for ploughing and trenching operations.

At the heart of the new ship are four Ulstein Bergen diesels in two father and son arrangements, with a total output of 27,400hp, turning two Ulstein propellers in fixed nozzles. At the bow there is a 1,620kW azimuthing thruster, together with a 1,200kW tunnel thruster, whilst aft two 833kW tunnel thrusters combine to give the vessel outstanding manoeuvring capabilities.

All of these devices, plus the Ulstein high lift rudders, are linked to a fully redundant Kongsberg Simrad SDP21 dynamic positioning system which is in turn interfaced with spotbeam, DGPS, tautwire, laser and HIPAP systems to ensure reliable and accurate station keeping. According to the owners, the vessel has a maximum free-running speed of 17 knots.

The vessel's considerable electrical

needs are met by two 3,000kVA shaft generators, and two Caterpillar driven generator sets of 715kW each.

Apart from the visually obvious helicopter pad built up over the bow and designed to accept a Super Puma AS332L2, *Far Sovereign* is dominated by a substantial A-frame at the stern, a cluster of cranes and other items of deck machinery ringing the aft working deck. With a 160-tonne SWL, the A-frame can be used for all manner of deepwater construction projects but most notably, as in the vessel's current operation, deploying EMC's pipe trenching and backfilling ploughs. These can be lowered over the pipe or trench with accurate positioning facilitated by real time sonar and video systems. The trenching plough is capable of lowering a pipeline of up to 1.4m diameter into a trench of up to 2m in depth. The devices are able to operate at depths of 400m and are controlled from a special dedicated operations room. Power and telemetry signals are passed from this cabin via an umbilical to a subsea electronics pod mounted on the ploughs.

A 100-tonne SWL heavy lift offshore crane is mounted on the port side of the aft deck while three smaller ones occupy positions at the other corners. The main winch has a maximum pull of 500 tonnes and houses up to 6,100m of 90mm diameter wire.

A 7m x 5m working moon pool is incorporated into the 680m² of high load

open deck space of the vessel's main deck and a removable modular ROV deployment deck can provide an additional 320m² of equipment storage room.

Main Particulars

Operator

Farstad Shipping, Norway

Builder

Langsten Slip, Norway

Length oa 85.20m

Breadth mld 20.50m

Draft (design) 7.00m

Main engines

4 x Ulstein Bergen diesels,
6,850hp each

Propulsion

2 x Ulstein 9275kW props in
nozzles

Bollard pull 280 tonnes

Speed (maximum) 17 knots

Accommodation 63

Crew 13

Equipment

Anschutz gyro compass and
autopilot, Furuno echosounder
and speed log, Taiyo VHF, Skanti
radio, Kongsberg Simrad SDP21
DP system



***GECO EAGLE*: seismic research vessel sets new standards**

Built for Schlumberger Geco-Prakla and subsequently despatched to work in Brazilian waters, the 95m long *Geco Eagle* - delivered in the first quarter of '99 - is probably the biggest and most advanced seismic research vessel in the world.

Designed by Skipsteknisk of Norway with interiors by civil architect Eva Eide, the vessel was built at the Mjellem & Karlsen (M&K) yard in Bergen as the latest in the builder's impressive portfolio

of research ships. Aside from its sheer size, the ship embodies a wealth of seismic towing technology.

Whilst dimensions of 94.80m x 24.00m are impressive, the ship's true size can only be gauged by taking into account its superstructure. This flares out, delta-shaped in plan, to a maximum breadth of 37.50m in order to provide the widest possible opening from which to deploy 8km-long seismic streamers.

Power is provided by a quartet of Wärtsilä 9L32 diesel generator sets, each of 4,050kW. These supply two 4,200kW Siemens IRN2803 electric motors, driving Wärtsilä Wichman SCV 95 reduction gears and 4,000mm diameter controllably pitch propellers inside fixed nozzles. This configuration gives the vessel a speed of around 16.25 knots.

To minimise noise and disturbance to underwater sensors, the screws are ensured

a smooth water flow by the twin skeg hull design aft. Manoeuvrability is enhanced by a 1,500hp, AR 80 azimuthing thruster and a 1,000hp, FU-80 transverse tunnel bow thruster, both from Brunvoll. Ulstein Group companies provided the steering gear and high lift rudders and the emergency electrical power is supplied by a Scania DSI 11 diesel/Stamford generation set of 260kW.

A large helicopter pad dominates the fore deck, ahead of the bridge and central accommodation block, which has facilities for 66 people. Side hangers beneath the bridge are used to house a Norsafe 885 Miriam lifeboat on the port side and a Norsafe 750 Magnum fast rescue boat to starboard. The latter is powered by a 230hp Yanmar engine with Hamilton 212 waterjet. Further aft on the starboard side, under the superstructure overhang, is an additional hanger for *Geco Bluebird*, a Norpower Malo 35 seismic cable workboat, powered by twin Cummins 6BTA diesels. All three vessels are launched and recovered by Vestdavis.

Other deck machinery, largely installed on the wide aft deck

above the streamer deck, includes two Aukra deck cranes and a number of Rapp Hydema winches.

Most of the electronic systems on board were supplied by the Kongsberg Group, owners of the brand names of Norcontrol, Simrad and Robertson. Equipment includes the vessel management system, radar/ECDIS Databridge, with two X-band and one S-band units, RGC12 gyro compass and two echosounders. The communications equipment is mainly of Sailor manufacture but includes a Nera Saturn Inmarsat B terminal. Vector-charting for the ECDIS is by C-Map.

Having an existing fleet of 27 vessels, Geco-Prakla, together with its parent Schlumberger, has gained a wealth of experience in seismic towing technology. This, combined with the expertise of Odin AS, the supplier of the complete seismic cable handling arrangement, has produced a system that can cater for up to 20 streamers.

Geco Eagle is the twenty-fifth research vessel built by M&K since the 1950s and follows two 90m vessels delivered to Eidesvik Survey in 1997.

Main Particulars

Owner

Seismic Shipping, Inc/Geco-Prakla

Builder

Mjøllem & Karlsen Verft, Norway

Designer

Skipsteknisk, Norway

Length oa 94.80m

Length bp 85.80m

Beam 24.00m

Beam (maximum aft) 37.00m

Propulsion

A diesel-electric propulsion system comprising: 4 x Wärtsilä 9L32 drive generators, 4,050kW each; 2 x 4,200kW Siemens IRN2803 electric motors; 2 x Wärtsilä Wichman SCV 95 reduction gears; 2 x 4,000mm CP propellers inside nozzles

Speed 16.25 knots

Classification

DNV +1A1, EO - ICE C - HELDK - SF

Equipment

Norsafe Magnum 750 MOB boat, Miriam 8.85m lifeboat, Norpower Malo 35 workboat, Norcontrol radar/ECDIS, Phillips DGPS, Simrad gyro and echosounders, Furuno logs, Aukra deck crane, Alfa Laval Nirex fresh water generator



The seismic vessel "Geco Eagle"
built by The M&K Group.
"Ship of the year award"
NOR-SHIPING 1999



MV "Veritas Viking II"
Delivered June 1999

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***HUMBER CALLISTO*: super star joins pilot fleet**

Associated British Ports at Hull, responsible for pilot services on the River Humber, has traditionally operated Scania powered twin-screw Nelsons and currently has a fleet of three with another in reserve. Tradition has, however, now been cast aside by Bevan Thomas, marine engineering manager of ABP Hull, with the delivery of *Humber Callisto*, a triple-engined waterjet driven vessel which, it is claimed, marks a watershed for pilot boat design and construction.

Designed by Alastair Cameron of Camarc Ltd and built on the Isle of Wight by Souter Marine, the 17.2m x 4.60m craft owes much in overall concept to a series of longer Rotterdam pilot boats from the same designer and built by Engelaer in Holland. Waterjet propulsion, advanced double chine hull, the bright livery and all-round fendering system have their roots in the Dutch boats but the three-engine/waterjet propulsion system is, according to Thomas, a first for a pilot boat.

Three Scania D114 engines, each developing 600hp at 2,100 rev/min, are located across the boat aft with the two wing engines sitting slightly forward of the centre unit. All three are direct coupled (no gearboxes) to Hamilton 362 waterjets using Centa flexible couplings and drive shafts. This arrangement gives a fully laden (two crew, ten passengers/pilots and 2,000 litres of fuel) continuous operating speed

of 32 knots and a top speed of 35 knots, well exceeding expectations. Commented Thomas: "It would have been difficult to reach these speeds by using two larger engines because of the increased weight of the engines and associated systems." A major advantage of waterjet propulsion is the outstanding manoeuvrability when approaching and leaving large ships in motion and the triple arrangement offers flexible operating speed alternatives.

Cameron's double chine hull has been proven on a number of vessels since its first development in the mid-eighties. It combines a number of advantages including finer entry in the forward sections, efficient spray rails to keep the deck dry and dampen motions in heavy seas (important attributes in a pilot boat) and a more seakindly shape in the forward slamming areas. A chine shape at the aft end also gives stability in following seas.

The wheelhouse/saloon is mounted on six anti-vibration mounts and to further reduce noise and vibration levels, Halyard Marine & Industrial was called in to design and manufacture a complete exhaust system with the task of reducing levels to around 75dB(A) in the deckhouse. The effect of the Halyard system, which incorporates modular high risers with high-temperature silicone bellows and specially designed in-line silencers, has been to bring the level down to 71 dB(A).

A major innovative feature of the boat is a stern mounted man overboard recovery device designed jointly by Cameron and Bevan. Using a small remotely controlled winch, a metal mesh and tubular scoop running in a channel over the stern can be lowered into the water to pick up the victim and raise him/her above deck in the prone position. The fendering system is also unique being a further development, by Camarc, of the original 'Popsafe' system used on the earlier Dutch boats. The basic principle of holding a tubular rubber fender in position around the hull by a tensioned wire rope is retained but the recess in which the tube sits is now formed from large sections of elastomer foam mouldings. Mechanical fastenings around the hull, and their weakening effect, are thus eliminated except where the tensioning points are located on the transom. A Hadrian pilot rail system is fitted to the foredeck.

The wheelhouse, which is air conditioned by a Crusair/Taylor Made Environmental system, provides excellent all-round visibility with an abundance of windows, supplied by Seaglaze Marine, especially in the vital overhead area. Three Hepworth pantograph wipers are fitted to the front screens whilst two pendulum systems keep the side screens clear.

Centrally located, the command position not only has wheel steering but also a



Left: the command position has wheel steering and Simrad Robertson follow-up lever.
Below: stern-mounted MOB device.

Simrad Robertson follow-up lever to port. On the same side is the Servowatch engine room monitoring screen. The Raytheon radar and plotter are to starboard, where they are also visible by the crewman/navigator who can grab the throttle and jet reversing bucket controls in an emergency. Directly ahead of the navigator is a Robertson AP9 Mk3 autopilot with the extensive Sailor radio (with DSC) fit-out to his left between the two Recaro N-Joy seats. The DGPS (again by Raytheon) along with depth, wind and speed indicators are ranged overhead. Behind the crew position are ten seats for the pilots, set out in forward facing pairs along each side. All the seating is equipped with safety belts. On the port side of the control console, steps lead down to the forecabin area which has a galley and twin berths.

Humber Callisto is impressive on virtually

every level from its sleek and attractive appearance to the thoughtfully placed interior handrails overhead along both sides of the aisle. The vessel is in active service whilst undergoing a stringent evaluation programme by all concerned over the coming months. If it meets with unanimous approval, ABP Hull will take out an option with Souter for a further two boats.



Main Particulars

Owner

ABP, UK

Builder

Souter Marine, UK

Designer

Camarc Ltd, UK

Length oa (excl. fenders) 17.20m

Length wl 14.46m

Beam mld 4.64m

Draft 0.80m

Main engines

3 x Scania D114 diesels, 600hp each

Propulsion

Hamilton 362 waterjets with Centa flexible couplings and drive shafts

Speed (service) 32 knots

Speed (maximum) 35 knots

Crew 2

Pilots/passengers 10

Equipment

Hadrian pilot rail system, Seaglaze windows and doors, Transit Super Nova 1400 pilot seats, Raytheon radar, plotter, DGPS and loudhailer, SG Brown Meridian gyro compass, Robertson AP 9 autopilot, Jotron Tron 30 EPIRBs, Sailor VHF, Servowatch monitoring and alarm systems, Halyard (M&I) exhaust systems, Hepworth pantograph and pendulum windscreen wipers, Crusair AC system



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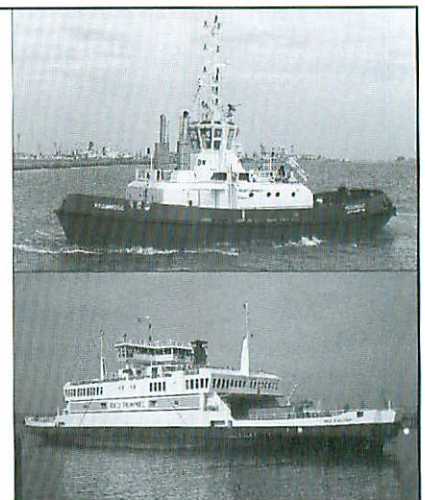
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RT INNOVATION CLASS: 'Rotor' tugs adopt unique tractor concept

One of the most intriguing tug concepts in recent years became reality in January '99 when *RT Innovation* and *RT Pioneer*, the first in a series of four, were delivered to Dutch operator Kotug. Designated 'Rotor' tugs, the initial duo - built in Spain at Astilleros Balenciaga - were soon to be followed by *RT Spirit* and *RT Magic*.

Conceived and patented by Ton Kooren, president of Rotterdam based Kotug - Sleepdienst Adriaan Kooren BV - the Rotor tug represents a new concept in tug design. At its heart, the concept is a unique tractor tug propulsion system utilising three main engines, each driving a fully azimuthing propulsion unit. Two propulsion units are located forward of the centreline, in the normal tractor configuration, with the third unit aft of the centreline replacing the traditional skeg. This seemingly complex arrangement offers a number of significant advantages, particularly in the close quarters ship-handling role.

The Rotor tug has the ability to turn

rapidly in its own length and apply almost all of its bollard pull in any direction - including sideways. The lack of a substantial skeg reduces resistance to turning and cuts down the influence of a ship's propeller wash when working in close proximity to large vessels underway. With 6,300hp installed in a 31m hull and a bollard pull of 75 tonnes, the Rotor tug is one of the most powerful and compact harbour tugs in existence. In fact, Kotug claims that if a breakdown occurs on one engine or propulsion system, the tug can remain operative with some 50 tonnes of bollard pull still available.

Construction of the series has involved three shipyards. The hull and all major steelwork for *RT Magic* was completed by Construcciones Navales P Freire SA in Vigo, Spain, and fit-out and completion undertaken by the Padmos Shipyard at Stellendam in Holland. *RT Spirit* was also ordered from Freire and handed over in February '99. Astilleros Balenciaga delivered *RT Innovation* and *RT Pioneer* as previously mentioned. All four tugs are

registered under the Dutch flag and classed GL+100A5 - "E TUG" +MC.E Aut.

Measuring 31.61m x 12.00m, each Rotor tug has a heavily built single chine steel hull and a semi-raised forecastle. No underwater skeg is fitted but some protection is provided to the propellers and for docking purposes by guard plates and struts. Three very small fins located under the stern enhance directional stability.

Three Caterpillar 3516B DT-TA main engines are installed, each with a maximum continuous rating of 2,100hp running at 1,600 rev/min. Each engine drives a Schottel SRP 1212 FP, fully azimuthing propulsion unit, incorporating a five-bladed fixed pitch propeller of 2.15m diameter and a Kort nozzle. Power is transmitted through Twin Disc Type 3000-2LD Marine Control Drive (MCD) units. The main engines are situated side by side across the engine room, with the outboard engines coupled to the forward Schottel units and the centre engine driving the aft unit. On trials the tugs produced a bollard pull ahead of 75 tonnes and 73.5 tonnes



astern, with a free-running speed of 12 knots.

Towing equipment is substantial and the after deck is dominated by a Ridderinkhof TW-H-300 'waterfall' type hydraulic main towing winch with three declutchable drums. The full width upper drum accommodates 650m of 56mm diameter steel wire rope, with a 20m Nylon (spring) of 100mm diameter. Each lower drum, used for ship-handling, carries 200m of 56mm steel wire rope, with a 10m Nylon spring and 48mm diameter steel wire pennant. On the foredeck is a Ridderinkhof AMW-H-236 combined towing winch and anchor windlass. The winch has a single drum, for a 200m towline with the necessary spring and pennant, two declutchable chain lifters and two fixed warping heads. Both towing winches have pneumatic controls and can be operated remotely from the wheelhouse or from small control consoles on deck. A Van de Graaf type GSH 100 quick release towing hook is fitted for secondary use.

The high rise Rotor tug wheelhouse has been designed according to ergonomic principles and affords excellent visibility. The tugmaster manoeuvres the tug from a central

position between two large consoles containing most of the propulsion, winch, communications and navigational equipment. The layout of the Schottel propulsion controls in the tugs is unique. Each propulsion unit has its own manual controller for thrust direction and propeller speed while a fourth 'Masterpilot' controller enables any combination of units to be directed in unison. The master can, therefore, choose the most suitable combination for the operation he is carrying out. In much the same way, the Anschutz Pilotstar D autopilot can be coupled to one, two or all three propulsion units. For example, at sea, the forward propulsion units are connected to the autopilot and the stern unit fixed in the straight-ahead position. All of the Kotug captains involved in the operation of the tugs were given extensive simulator training by Marine Safety Rotterdam BV, along with pilots and many others involved with the introduction of the new vessels.

Crew accommodation is fitted out to a very high standard. The tugs will normally work in harbour with a crew of three, with additional manning for seagoing operations. Provision is made for a maximum of seven personnel, in three single and two twin-berth cabins.

Above: a truly impressive wheelhouse. Below left: three propulsion units, two forward and one aft of centre. Below right: a 'waterfall' type winch on the aft deck.

Main Particulars

Owner & Designer

Kotug, The Netherlands

Builders

Ast. Balenciaga, Spain, P. Freire SA, Spain and Padmos Shipyard, The Netherlands

Length oa 31.63m

Beam mld 12.00m

Draft 5.90m

Main engines

3 x Caterpillar 3516B DITA diesels, 2,100hp each

Propulsion

3 x Schottel SRP 1212 Rudderpropellers driven via Twin Disc 3000-2LD MCD clutches

Speed 12.5 knots

Bollard pull ahead 76 tonnes

Bollard pull astern 73.5 tonnes

Classification

GL +100A5 - "E TUG" +MC.E Aut

Equipment

Ridderinkhof winches/anchor windlass, Van de Graaf towing hook, Anschutz autopilot, Skipper echosounder, Skanti VHF's, JRC Navtex, CSI bridgewatch system, Azcue hydraulic winch pump, K&R fire pump, Unitor fire-fighting monitor





MARCI MORAN: first of four for naval duties

Four new ASD tugs ordered by US operator Moran Towing are now in service at two Military Sealift Command bases in the Norfolk, Virginia area. The prestigious design and build contract was won by Washburn & Doughty of East Boothbay in Maine, respected commercial vessel builders that had surprisingly little experience in the tug sector.

The key to success in winning the order undoubtedly lies with the delivery of *Fort Bragg*, a 27.84m x 9.75m Z-drive reverse tractor designed by Bruce Washburn and built at his yard in 1998. This in turn was inspired by a twin-screw tug design of 1991 but, aside from this, the Washburn portfolio had little else to help thwart intense competition from other builders.

The lead vessel for Moran - *Marci Moran* - measures 28.04m x 9.75m and has a draft of 4.2m. Power is provided by a pair of EMD-645-E2 diesels, each developing 2,100hp at 900 rev/min. These drive stern-mounted Ulstein type 1350H Z-drives with stainless steel propellers turning inside stainless lined Kort nozzles. Twin Detroit 4-71 gensets with a total output of 100kW supply auxiliary power. A third Detroit diesel engine - an 8V71T - rated 280hp at 1,800 rev/min is included

in the package to turn an Aurora fire pump, which has a discharge rate of 3,000 US galls/min. This feeds two Skum fire-fighting monitors located in the forward starboard and aft port quarters of the bridge deck.

The compact wheelhouse, accessed internally to port and externally on the inside of the starboard exhaust stack, has parallel consoles forward either side of the helmsman's seat. The chunky rectangular exhaust stacks scarcely obstruct good all-round views and all navigational equipment - including Furuno GPS, depth sounder and radar - are embedded in the control consoles.

Crew accommodation is arranged in the deckhouse with two crew cabins, each with double bunk beds forward of the engine room and a shared cabin for captain and mate to port. A roomy combined galley, dinette and lounge section takes up most of the starboard side. The dining area features a U-shaped seating arrangement, which is sufficient to seat all crew members, and continuous seating along the forward and side bulkheads for up to ten extra people. This additional space will be used when vessel crews are being ferried.

Main Particulars

Owner

Moran Towing, USA

Builder

Washburn & Doughty, USA

Designer

Bruce Washburn

Length oa 28.04m

Beam 9.75m

Draft 4.20m

Main engines

2 x EMD-645-E2 diesels,
2,100hp each

Propulsion

2 x Ulstein 1350H Z-drives with
stainless steel propellers turning
inside Kort nozzles

Auxiliary power

2 x Detroit 4-71 gensets, 50kW
each

Equipment

Skum fire-fighting monitors,
Aurora fire pump, Furuno GPS,
depth sounder and radar, ICOM
VHFs, Schuyler soft loop
fendering, MARCO electric
capstans



MARK 10 LCU: prototypes in test for 2000

Two prototype vessels of Mark 10 LCU (landing craft utility) type are undergoing extensive operational trials during 2000 with the UK Royal Navy. The duo form part of a £35 million ten-vessel design and build contract placed with the Land & Sea Systems Division of British Aerospace and are being built under sub-contract at Ailsa-Troon Shipyard on the Scottish west coast.

The vessels form the initial stage of the Navy's programme to improve its amphibious capability and are the first to have a ro-ro facility with electrohydraulically operated bow and stern ramps. The forward ramp is wide enough to take a Challenger tank whilst aft access, restricted by the offset wheelhouse, can accept most normal vehicles. Both ramps were supplied by Transmarine.

With a displacement of 240 tonnes, the LCUs measure 29.8m overall with a beam of 7.4m and a draft of 1.5m. Two completely self-contained engine rooms each house a 400kW MAN D2840LE diesel and one of two differently sized Cummins/Newage generator sets. This arrangement ensures that the craft are still operational in the event of a single engine room sustaining damage. The larger generator is an air-cooled set having an

output of 100kW and the other, a water-cooled unit used for normal cruising, is of 50kW. Selected in order to provide trouble-free operation when the craft is beached, the air-cooled arrangement provides power to three hydraulic power packs for the ramp and keel winches, plus supplying a bow thruster and under deck tract heating.

Both main engines drive a model SPJ82 PumpJet mounted at the stern. A similar, but smaller, electrically driven SPJ15RD PumpJet is fitted at the bow. The units are flush mounted to prevent damage when grounding.

Jigger winches raise and lower the bow and stern ramps, each of which has an independent electrohydraulic power pack. The third pack is dedicated to the keel winch but interconnection is possible in an emergency.

Designed to operate from the new landing platform dock vessels, *HMS Bulwark* and *HMS Albion*, currently under construction at the Marconi Marine yard in Barrow-in-Furness, the LCUs are capable of operating in a combat situation worldwide. Each can carry 120 fully equipped troops (with skis and two over-snow vehicles), four high-sided vehicles or a main battle tank. The first two vessels were built to schedule within 14 months and were handed over simultaneously.

Main Particulars

Operator

Royal Navy, UK

Builder

Ailsa-Troon, UK

Designer

British Aerospace Land & Sea Systems Division

Length oa 29.8m

Beam 7.4m

Draft 1.5m

Displacement 240t

Capacity 120 troops

Main engines

2 x MAN D2840LE diesels, 400kW each

Main generators

Cummins/Newage gen sets, 100kW and 50kW

Propulsion

Type SPJ82 Schottel PumpJet at the stern and SPJ15RD PumpJet at the bow



MIDNIGHT ARROW: Multi-role OSV starts work in Mexican Gulf

The 60m long survey/ROV/offshore support vessel *Midnight Arrow* has recently begun a long term charter in the Gulf of Mexico with Torch, Inc. The vessel was built by the revitalised Balenciaga yard to a Cintranaval design for Dublin-based Adams Vessels (Bilbao) Ltd, a wholly owned subsidiary of Khalifa A Algosaihi Diving & Marine Services of Saudi Arabia.

As towing is not a prerequisite, power is comparatively low for an OSV of 60.00m x 13.30m x 3.80m. Two Caterpillar 3512TA diesels, each developing 1,297hp at 1,800 rev/min, are installed under the centre of the deck abaft the moonpool. These turn stern-mounted Schottel twin propeller units, type STP 1010, through flexible couplings and inclined shafts. By using these 'dual-prop' units, it has been possible to minimise the diameter and hence, the operating draft. At the bow, two 350kW Schottel type STT 170 transverse thrusters are fitted.

To meet the vessel's high electrical demand, four Caterpillar driven generator sets are located in a separate auxiliary room adjacent to the main switchboard compartment. Three, arranged for synchronous operation, are 425kW, 3412 units whilst a fourth, solely dedicated to ROV operations, is a 3408TA producing 370kW. A 64kW emergency set is located on the forecastle deck.

Diving support, pipeline construction and maintenance are the vessel's primary tasks and, as much manpower is needed

to perform these operations, the vessel has accommodation for 52 persons. Most of the quarters are on the main deck and are notable for their spaciousness. On the upper deck are two four-berth cabins, separate ROV and project offices, comfortable day room, mess and galley. The mess has seating for 42 at seven tables and is equipped with a coffee machine. Forward of the galley are extensive dry and cold stores. The captain and chief engineer each have extensive 'suites' at the forward end of the forecastle deck, together with other officer cabins, guest accommodation and the ship's office.

With dimensions of 14m x 8m, the wheelhouse is large by any standards. Walk-round consoles fore and aft duplicate most controls (Schottel) and communication systems (Sailor) whilst the unit facing ahead is flanked by two Furuno radars. Just aft of the port wheelhouse door is the Furuno Inmarsat B equipment, alongside a battery of fax machines. A Sailor 'compact' GMDSS system backs onto a large chart table. Other electronic items include a Furuno echosounder and a Raytheon GPS.

A prominent feature on the starboard aft quarter is the dual screen Nautronix DPS2 dynamic positioning system which can accept DGPS, hydro-acoustic and taut wire references. According to Balenciaga, this is the first Spanish built vessel of this type to be DP equipped and on trials, in 30-knot winds and 3m seas, the ship was

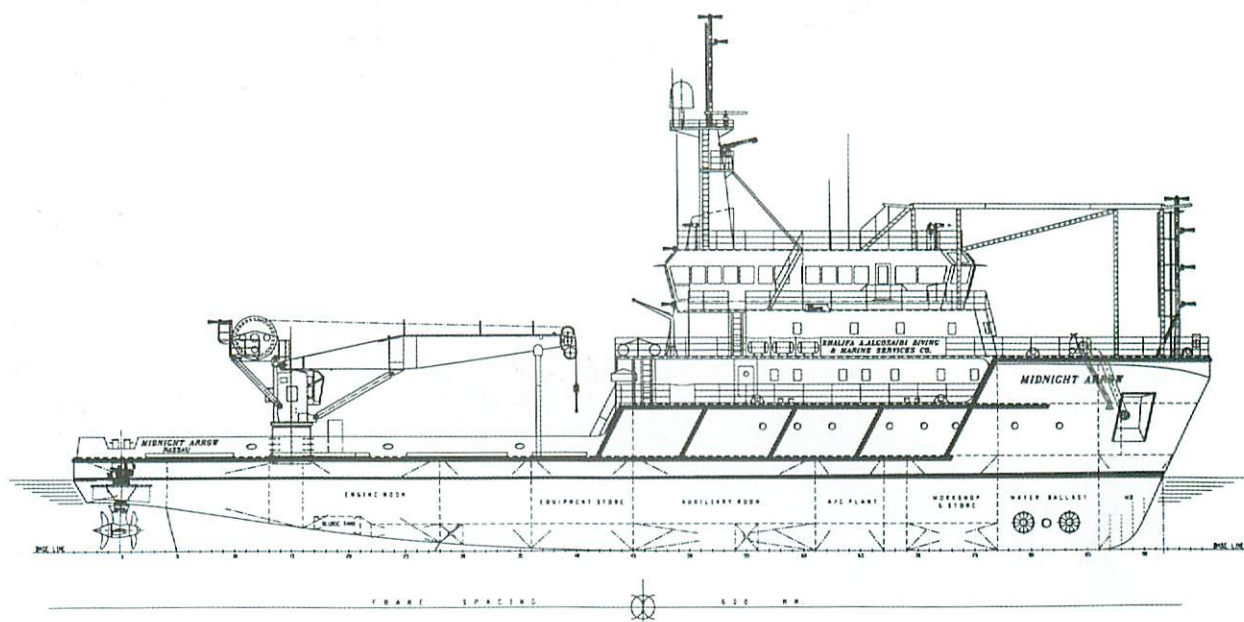
able to maintain station to within 400mm.

A companionway at the aft end of the wheelhouse leads to the roof and on up to the helicopter deck. This is an aluminium structure supported on steel uprights and is protected by two fire-fighting monitors mounted on a platform high on the mast.

Dominating the timber clad aft deck, which has an area of 310m², is a Norlift crane with a 45 ton SWL at 10m outreach. Unusually, this is located near to the transom, which has a wide gate. Other deck machinery items include a pair of Ridderinkhoff capstans and a forward windlass. A Zodiac rigid inflatable,



MIDNIGHT ARROW



powered by a 70hp Tohatsu outboard, is swung from the aft end of the bridge deck.

To ABS classification requirements and supervision, the vessel has been completed in under 15 months from the signing of the contract and to the entire satisfaction of the customer, who may order a second vessel.

Now under a new management team, Balenciaga has, over the last three years, entered a period of greatly improved profitability with the result that an extensive investment programme is in

progress. Already a large dedicated pipe shop has been constructed and a new NC bending machine installed. This has facilitated work on 2,000m of pipe installed on *Midnight Arrow*. Two more workshops (52m x 18m and 70m x 15m) are to be added before mid-2000, together with a total of 215m of deep water quay. This latter facility will result from a dredging exercise - the final stage in a marsh/mud reclamation project along two sides of the original yard perimeter.

Main Particulars

Owner

Adams Vessels (Bilbao) Ltd,
Spain

Builder

Balenciaga, Spain

Designer

Cintraval, Spain

Length oa 60.00m

Beam 13.30m

Draft 3.80m

Main engines

2 x Caterpillar 3512TA diesels,
1,297hp each

Propulsion

2 x stern-mounted Schottel type
STP1010 propellers driven via
flexible couplings and inclined
shafts. 2 x STT 170 transverse
thrusters at the bow

Accommodation 52

Equipment

Norlift crane, Ridderinkhoff
capstans and windlass, Zodiac
RIB, Furuno radars and
echosounder, Sailor GMDSS
and VHF's, Raytheon GPS,
Nautronix DPS2 dynamic
positioning system



NANUQ & TAN'ERLIQ: powerful tractors for tanker escort

Two of the most powerful Voith Schneider tractor tugs in the world, *Nanuq* and *Tan'erliq* (Alaskan words for 'polar bear' and 'black bear' respectively) entered service in the first half of '98.

Designed by Seattle based naval architects Guido Perla & Associates to carry out tanker escort duties in Valdez Harbour and Prince William Sound, each vessel is also fully equipped for ship handling, fire-fighting, emergency response and anti-pollution work. Owned by Crowley Marine Services (CMS) of Seattle, the tugs are on lease to the Alyeska Pipeline Service Company.

Built at a contract price of around US\$31 million by Dakota Creek Industries in Anacortes, Washington, the 46.63m x 14.63m sisters have all-welded steel structures with transverse framing and wing tanks, strengthened for operation in ice. Tank spaces are provided for 123,280

US gallons of fuel, 17,800 gallons of fresh water and 3,000 gallons of lubricating oil.

The hull design incorporates a prominent forecastle, extending almost half the length of the vessel, and a streamlined configuration. A substantial 'breakwater' located on both after deck and forecastle is essential for safety in arduous working conditions. Heavy moulded rubber fendering extends around the entire hull, with an additional stern fender fitted for push-pull operation. The underwater hull is shaped to give good handling characteristics while operating stern first, and incorporates the usual hydrodynamic skeg.

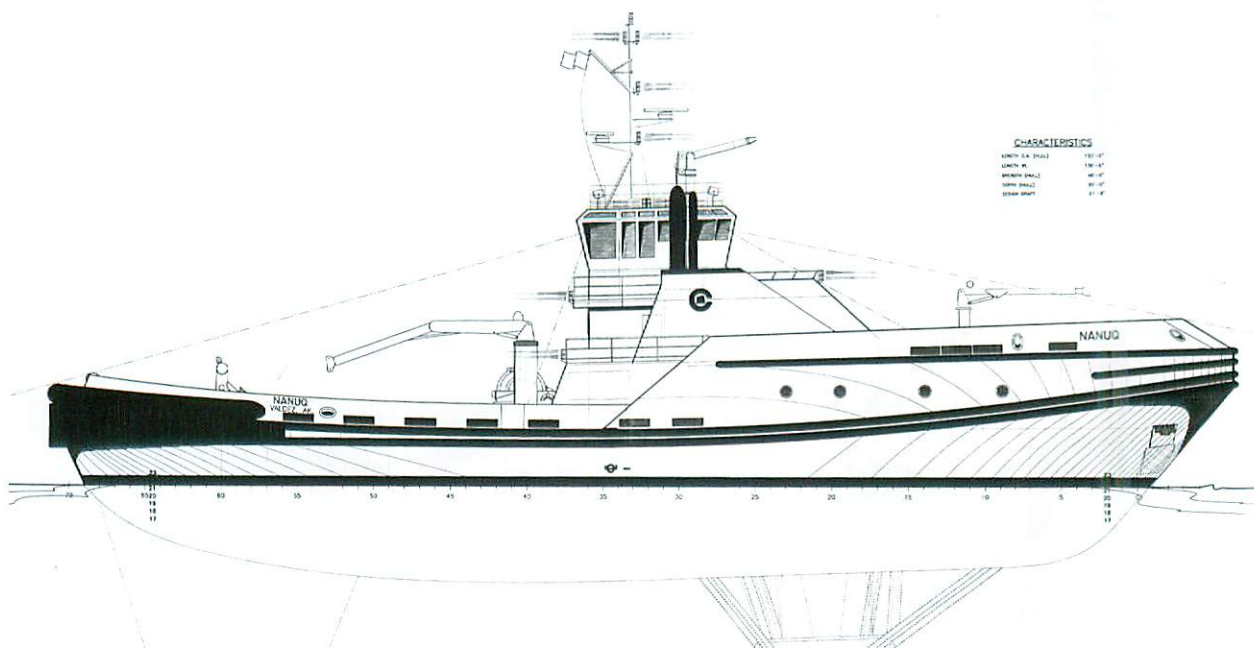
Two Caterpillar 3612-B turbocharged, four-stroke diesel main engines, produce a total of 10,192hp at maximum continuous power and drive a pair of Voith Schneider Model 36G11/270 cycloidal propulsion units. This arrangement ensures a static

bollard pull of some 104 tons and a free-running speed of 15 knots.

On deck, the towing gear comprises a Markey DTSDS-62 split drum towing winch, the usual 'staple' fairlead and a further lead in the after rail. The main towline is 1000ft long 10 inch circumference D-19 SWR, with a pennant of 15 inch Nylon, 200ft in length. For escort work and ship-handling the tugs are equipped with a 1000ft towline of Spectron-12 Plus, 12 strand braided, man-made fibre rope of 10 inch circumference.

Three hydraulic deck cranes, supplied by North American Marine Cranes, are installed (one forward and two aft) to handle anti-pollution booms, stores and the onboard workboats.

Two Skum fire-fighting monitors are located above the wheelhouse roof, each with a capacity of 5,300 galls/min of water. Two pumps supply the fire-fighting system



with a total capacity of 13,200 galls/min. Some 9,400 gallons of fire-fighting foam is carried on board in dedicated tanks.

The centrally positioned wheelhouse has all-round vision, impaired only by the stainless steel uptakes of the engine exhausts. A substantial communications and navaid package is installed and

includes: two Furuno 2110 ARPA radars; Robertson gyrocompass and autopilot; Trimble NT 200D GPS and electronic chart display; and Furuno LS 6000 depth sounder. Radio equipment is to GMDSS standard, with three SEA 156 VHF sets, Sat Com and a Furuno weather fax.

Main Particulars

Owner

Crowley Marine Services, USA

Builder

Dakota Creek Industries, USA

Designer

Guido Perla & Associates, USA

Length oa 46.63m

Beam oa 14.63m

Draft (loaded) 6.71m

Speed 15 knots

Bollard pull 95.5 tonnes

Main Power

2 x Caterpillar 3612-B diesels,
5,096hp each

Propulsion

2 x Voith Schneider Type 36G11/
270 cycloidal propulsors

Classification

ABS+A1 Towing Service, +AMS,
+ABCU, A1 Fire-fighting, non-
dedicated OSRV & FFVI Class 1

Equipment

Markey DYSD-62 towing winch,
Skum fire-fighting monitors,
Furuno radars, depth sounder
and weather fax, Robertson
gyrocompass and autopilot,
Trimble GPS



NEELTJE P: first in Towmaster series

The twin screw tug *Neeltje P* has become the first of a new class of vessel designated 'Towmaster' due to its patented rudder system. Designed in the UK by Burness, Corlett & Partners (BCP), the tug - along with two subsequent identical sisters - was built in Santiago at Astilleros Detroit Chile for Terminales Maracaibo in Venezuela.

Neeltje P measures 30.40m x 10.0m with a draft aft of 4.0m and has a hull based upon BCP's much proven and easy to build double chine form. It has a transom stern and well-rounded bow with a half-height forecastle to provide freeboard at the bow as well as space below for crew accommodation. Due to low profile exhausts, positioned well aft, the wheelhouse has exceptionally good visibility for manoeuvring and safely controlling deck machinery.

The vessel is powered by a pair of Detroit Diesel 12V-4000 engines, developing a total of 3,384kW (4,600hp) at 2,100 rev/min, driving 2,600mm diameter propellers with fixed nozzles. Bollard pulls in excess of 60 tonnes ahead and 41 tonnes astern were achieved on trials, together with a 13.6-knot maximum speed. Two

Detroit Diesel/Marathon generator sets, each of 56kW, supply the electrical power and a 500hp Detroit Diesel 8V71 drives an 800m³/h fire pump.

As well as providing detailed design plans for the tug, BCP designed the Towmaster system. For *Neeltje P*, the most recently developed version of triple high lift rudder system was fitted to both Kort nozzles. This arrangement provides almost pure side thrust when the rudders reach their maximum design helm angle. A Kobelt steering system gives both combined and independent control.

A single, a double and two triple berth cabins are situated forward of the engine room whilst the captain's cabin is on the forecastle deck alongside a spacious mess and separate galley. The wheelhouse has a T-shaped floor plan with a centrally located helm seat. A chart table is to port and a Furuno GMDSS station to starboard. Furuno also supplied the radar, sounder and GPS. Deck machinery items include a Kraaijeveld main towing winch on the aft deck, with 80 tons brake power and capacity for 600m of 50mm diameter wire, a 60-ton Mampaey towing hook and a NETEC anchor winch.

Main Particulars

Owner

Terminales Maracaibo, Venezuela

Builder

Astilleros Detroit Chile

Designer

Burness, Corlett & Partners, UK

Length oa 30.40m

Beam oa 10.00m

Draft aft 4.00m

Main engines

2 x Detroit Diesel 12V-4000 diesels,
1,692kW each

Propulsion

2 x fixed pitch propellers in Kort
nozzles, each fitted with triple
Towmaster high lift rudders

Speed 11.9 knots

Bollard pull (ahead) 60 tonnes

Bollard pull (astern) 41 tonnes

Equipment

Kobelt steering system, Kraaijeveld
towing winch, Mampaey towing hook,
NETEC anchor winch, Furuno radar,
sounder, GPS and GMDSS system



PACIFIC SUPPLIER: Ulstein design modified for extra flexibility

Pacific Supplier, the first of a series of four 58.03m x 15.00m sister ships, joined the fleet of Singapore operator Swire Pacific Offshore in early '99. Based upon an Ulstein UT 719 design but modified in accordance with the owner's instructions by the Japanese builder Imamura Shipyard, the vessel has greater capacity and flexibility for cargo carrying than standard UT 719 ships.

Traditional in terms of offshore layout, with forward superstructure, large aft deck and forecastle, *Pacific Supplier* has a top speed of 14 knots, service speed of 10 knots (burning an economic seven tonnes of fuel per day) and a bollard pull of 68 tonnes.

Propulsion is by means of twin Ulstein Bergen KRMB-9 main diesels, each developing 2,005kW at 825 rev/min, driving CP propellers in Kort nozzles. A trio of 7,000hp Ulstein 150TV electrically-driven tunnel thrusters are fitted, two at the bow and one at the stern. Enhanced manoeuvrability is ensured by incorporation of an Ulstein Poscon joystick, which is integrated to the propellers, engines, rudders and side thrusters and interfaced to the Tokimec gyrocompass and Simrad Robertson AP9 Mk3 autopilot.

Tanks for mud, brine and cement have been arranged, along with fuel oil, potable water, ballast and lube oil tanks, to allow a variety of products to be carried. All are centrally controlled by a tanktender system to meet specific customer needs. The tanks

for liquid mud, for example, have sloping bottoms and are free of internal stiffening to optimise cargo out-turn and assist cleaning.

A prominent feature of the aft deck is a totally enclosed winch house containing a remotely controlled Ulstein Brattvaag SL 150-B2T low pressure hydraulic winch, which has a brake holding load of 250 tonnes and a capacity of 1,400m of 56mm wire. Remote TV cameras in the winch-house area and monitors on the bridge ensure that a close watch can be kept on this potentially hazardous operation whilst minimising exposure to danger.

Other deck machinery includes two rig chain lockers, a 300-tonne SWL Karm shark's jaw, two tugger winches, a combined windlass anchor winch, two capstans, towing pins and stern roller. A deck crane is positioned at the aft end of the forecastle deck, along with a dedicated davit for launching a 25-knot Schat-Harding MOB boat, fitted with inboard diesel and waterjet.

Accommodation is provided for 24 with the mess, day room and galley on main deck and sleeping quarters and ship's office on the forecastle deck. The wheelhouse has fore and aft consoles and radio officer's position to starboard. Electronic equipment includes two Furuno radars and GPS, Skipper echosounder, Sailor watch receiver and Area 3 GMDSS unit.

Two months after *Pacific Supplier*

commenced operation in Australian waters, the second in the series, *Pacific Supporter*, was delivered. *Pacific Searcher* and *Pacific Server* were delivered in the final quarter of '99.

Main Particulars

Owner

Swire Pacific Offshore, Singapore

Builder

Imamura Shipyard, Japan

Designer

Ulstein & Swire Pacific

Length oa 58.03m

Beam 15.00m

Draft (summer) 4.72m

Main engines

2 x Ulstein Bergen KRMB-9 diesels, 2,755hp each

Propulsion

2 x CPPs in Kort nozzles

Auxiliary power

3 x Yanmar gensets, 2 x Taiyo shaft generators

Speed 14 knots

Bollard pull 68 tonnes



REDBUS 520: Cat style craft folds for road transport

A uniquely constructed foldable catamaran style workboat from newly formed company Redbus Workboats can be adapted for a huge range of work and leisure activities. Designated Redbus 520, the craft can be launched and recovered from a dedicated trailer in a matter of minutes making it ideal for fast response situations such as fire-fighting and pollution control.

Key to the innovative design is a patented system for locking the two GRP multi-section hulls to the bridge deck and, hence, unfolding the boat for operation. On recovery, the hulls are easily detached and folded so that the boat can be towed away.

Measuring 5.2m x 3.5m, the vessel is claimed to be highly stable in open sea and sheltered waters and offers more than 13m² of useable deck space, equivalent to that of many larger workboats of 9m or more. The central bridge deck is made from steel with a marine-ply 'non-slip' surface through which equipment may be secured. If required, the deck can be trap-doored to permit through-deck winching of up to 1.6 tonnes and can be easily adapted to incorporate securing points for onboard equipment.

Redbus 520 is recognised by the Marine & Coastguard Agency (MCA) as meeting the requirements of the Code of Practice

for Small Workboats & Pilot Boats for payloads of up to 1.6 tonnes. This represents a major advantage when compared to many RIBs and other vessels. Also of advantage are the different power and propulsion options of diesel or petrol engines. Redbus Workboats will supply and fit engines of customer choice, together with steering gear and console. As a guide, a four-stroke engine should ensure a speed of 20 knots for an unladen craft. The power range spans 27-120hp depending upon application.

Other bonuses are good manoeuvrability and shallow draft, which makes the Redbus 520 ideal for operations within the confines of harbours and marinas. Draft can be as little as 30cm, allowing activities such as structural surveys and repairs to be conducted with ease.

Commenting on the launch of the craft, Keith Lewis, managing director of Redbus Workboats, said: "The stability, strength and loading capacity of the Redbus 520 creates unique opportunities within many marine sectors, where for many years conventional boats have provided only limited capability. The Redbus 520's ability to be stored and transported on land gives an added dimension to the boat's flexibility. The Redbus 520 brings together the very best of British innovation with the

finest in British engineering."

To date, as evidence of its versatility, the 520 has been marketed most specifically to harbours, fire-fighters and pollution control experts. Leading pollution and environmental control company, Vikoma International, has purchased the vessel as part of its integrated, rapid response capability and UK fire crews have recognised the value of the craft's shallow draft. Utilising the easy-access ramp has enabled firemen to easily launch modern roll-on roll-off fire-fighting equipment.

Main Particulars

Builder

Redbus Workboats, UK

Length oa 5.2m

Beam 3.5m

Weight 960kg

(including 50hp 4-stroke engine)

Payload 1,600kg

Free deck space 13m²

Speed 20 knots

Main engines

Diesel or petrol

engines 27-120hp



RIVAGE ST MARTIN: first of new style 'boutique' cruisers

Australian shipbuilder, Austal Ships, contends that there is a growing need for a product that sits between the full-size cruise ship and the luxury charter yacht, offering a high level of exclusivity but at a price approaching that of the traditional cruising vessel. The yard has named this the 'boutique' cruise market and believes the concept works best with up to around 100 passengers.

Rivage St Martin, at 60m x 15m, is the first example built specifically for this new market and is currently cruising in the Caribbean. An abundance of luxurious features have been incorporated to cater to the comfort and relaxation of 80 passengers.

The vessel is an all-aluminium catamaran with semi-SWATH hulls which, combined with the ship's wide beam, provides excellent stability and a seakindly motion. A maximum draft of 2.1m is particularly suitable for a vessel that needs to moor close inshore in natural scenic lagoons rather than at busy ports or out at sea.

A service speed of 14.5 knots is derived from a twin MTU 12V 2000 M70 diesel arrangement, each unit developing 790kW at 2,100 rev/min, turning fixed pitch propellers through ZF BW 256 gearboxes. This speed will enable the vessel to arrive at a different destination each morning after only three hours sailing during the night. The operator, Rivages Crossieres, the Caribbean arm of leading French

maritime group, Compagnie Chambon, intends to run the vessel on a seven-day cycle, the final day being the changeover at home port St Martin.

The luxury one-class vessel features a very high standard of interior finish with 34 fully carpeted cabins on three decks - each with an exterior view. Single beds in the 16 two-berth cabins can be linked to create double berths whilst four cabins are capable of accepting two extra persons and four more have a single additional berth. All have private en-suite facilities, controllable air conditioning, safety deposit box and TV monitors connected to the ship's video system.

Central reception is situated on the main deck, along with a souvenir shop and 80-seat restaurant. The gourmet galley provides a wide range of French cuisine. A bar and lounge on the upper deck (again with seating for 80) adjoins a dance floor with disc-jockey station. There are also smaller bars by the saltwater swimming pool and on the sun deck.

For those guests seeking more action than lounging and drinking, a watersports boat is located on the upper deck and stern platforms ease the use of a wide range of equipment including water skis, snorkel sets, sailboards and sea kayaks. Three large rigid inflatable tenders are also provided for swift transfer to and from the shore.

Rivage St Martin operates with a crew of 20 (eight officers and ratings plus 12 hotel and catering staff). The captain and

chief engineer have accommodation abaft the bridge whilst the remaining crew quarters are to be found in the forward sections of each hull. Crew amenities include a mess area and a lounge fully equipped with entertainment systems.

Main Particulars

Owner

Rivages Crossieres, Caribbean

Builder

Austal Ships, Australia

Length oa 60.00m

Beam 15.00m

Draft Maximum 2.1m

Main engines

2 x MTU 12V 2000 M70 diesels,
790kW each

Propulsion

Fixed pitch propellers running via
ZF BW 256 gearboxes

Speed (service) 14.5 knots

Passengers 80

Crew 20

Classification

BV X 1 3/3E Coastal water 50nm
from port of refuge +MACH AUT
- MS, F



RIVER RUNNER: Rotterdam takes six for ferry service

Officially launched on 1st November '99, the 'Nieuwe wegen over water' project has entered into its first phase of developing a low-wash mass transit ferry service in The Netherlands.

Promoting waterbus transport as an efficient alternative to the use of congested road and rail networks, the service utilises four River Runner 150 ferries built in Holland by Damen Shipyards under a licence agreement with designers and builders NQEA of Australia. The latter also received a contract to build two of the larger Type 200 vessels in Australia for the same project.

With a view to making Rotterdam more accessible by water, two companies were formed to operate ferry services - High Speed Ferries BV and Waterbus BV. Waterbus BV represents a joint venture between ZWN and Stadsvervoer and will provide a link with the Drechtoever towns of Dordrecht, Papendrecht and Zwijndrecht every 15 minutes with a connection between Dordrecht, Sliedrecht, Mendrick Ido Ambacht and Albasserdam every 30 minutes. All towns involved have been provided with a purpose-built covered floating berth, located adjacent to car parking and bus interchange. Self-service ticketing is provided on three floating berths.

At 30.50m x 7.00m, each ferry is

powered by twin Caterpillar 3196 B DI-TA diesels driving conventional bronze five-bladed propellers via Twin Disc gearboxes. Two hydraulically driven bowthrusters are fitted. Contract performance was achieved at 80 per cent engine power. Aluminium hull fabrication was undertaken at Bayards Aluminium Constructie BV in Nieuw Zekkerland and the FRP sandwich superstructure at Damen.

Like the larger Type 200, the 150 is classed by DNV and has been designed and built with environmental limits on fuel consumption, emissions, noise and wash. Damen has reported that the measured wash on the Waterbus style River Runner is less than required in the contract and less than the standard set by the government. The 150 version caters for 130 passengers (50 standing and 80 in aircraft style seats) and can carry 60 bicycles. A special folding type gangway system is incorporated to facilitate swift boarding.

The Australian built craft are of River Runner 200 type, measure 37.00m x 8.00m and can accommodate 150 passengers and 20 bicycles. Powered by Deutz engines, each has a service speed of 30 knots and operates on a twice-hourly service, with a one-way trip between Dordrecht and Rotterdam taking around 25 minutes.

Both Damen and NQEA see other

opportunities in Europe for similar operations. A further River Runner is currently under construction and is due for delivery at the end of 2000 to another Dutch owner.

Main Particulars

Owner

Waterbus BV, Holland

Builder

Damen Shipyards, Holland

Designer

NQEA, Australia

Length mld 30.50m

Beam mld 7.00m

Depth at sides 2.22m

Draft aft 1.32m

Main engines

2 x Caterpillar 3196 B DI-TA diesels, 330bkw each

Propulsion

2 x bronze 5-bladed propellers, hydraulically-driven bowthruster

Speed (cruising) 20 knots

Speed (maximum) ... 22.70 knots

Passengers 130

Crew 2-3



ROEBUCK BAY: first of class for Australian Customs

The first of eight 35m Bay Class aluminium patrol boats, built at Austal Ships for the Australian Customs Service National Marine fleet, was deployed in February '99. Named *Roebuck Bay*, the vessel has been designed for a wide variety of functions including interception of unauthorised craft, smugglers and illegal immigrants and scientific research. Deliveries of two further vessels - *Holdfast Bay* and *Botany Bay* - followed, according to schedule, in August '99 and another pair will enter service as *Significant Small Craft* goes to press. The remaining three will be completed by August 2000.

Roebuck Bay has a range of 1,000nm and is capable of operating anywhere around Australia's 37,000km coastline and out to the edge of the EEZ, with a 12-man crew expected to undertake 28-day missions.

The semi-displacement hullform has a fine waterline entry designed for minimum resistance at 20 to 22 knots whilst affording maximum comfort in a seaway. As part of the tender process, the hull was tank tested to compare vertical accelerations with existing vessels in the customs fleet. Streamline tests were also carried out to align the bilge keels and ride control fins with the flow along the hull body. Since the vessel is expected to spend time either stationary or at slow speeds in beam seas, passive roll stabilisation has been

enhanced by adding shaft support skegs and bilge keels. For roll and pitch control, an Austal active ride control system was fitted comprising rotating cantilevered fins amidships and flaps aft.

Power is derived from a pair of MTU 16V M70 diesels, each developing 1,050kW at 2,100 rev/min, driving 1,150mm diameter four-bladed Veem propellers through Reintjes gearboxes. This configuration gave a trials performance of 20.5 knots at 80 per cent MCR in Sea State 3 - half a knot above the contract requirement. Electrical power generation is provided by two 135kW Cummins driven sets, each of which is individually able to supply the entire demand of the vessel with a 10 per cent reserve. A bow thruster is also fitted.

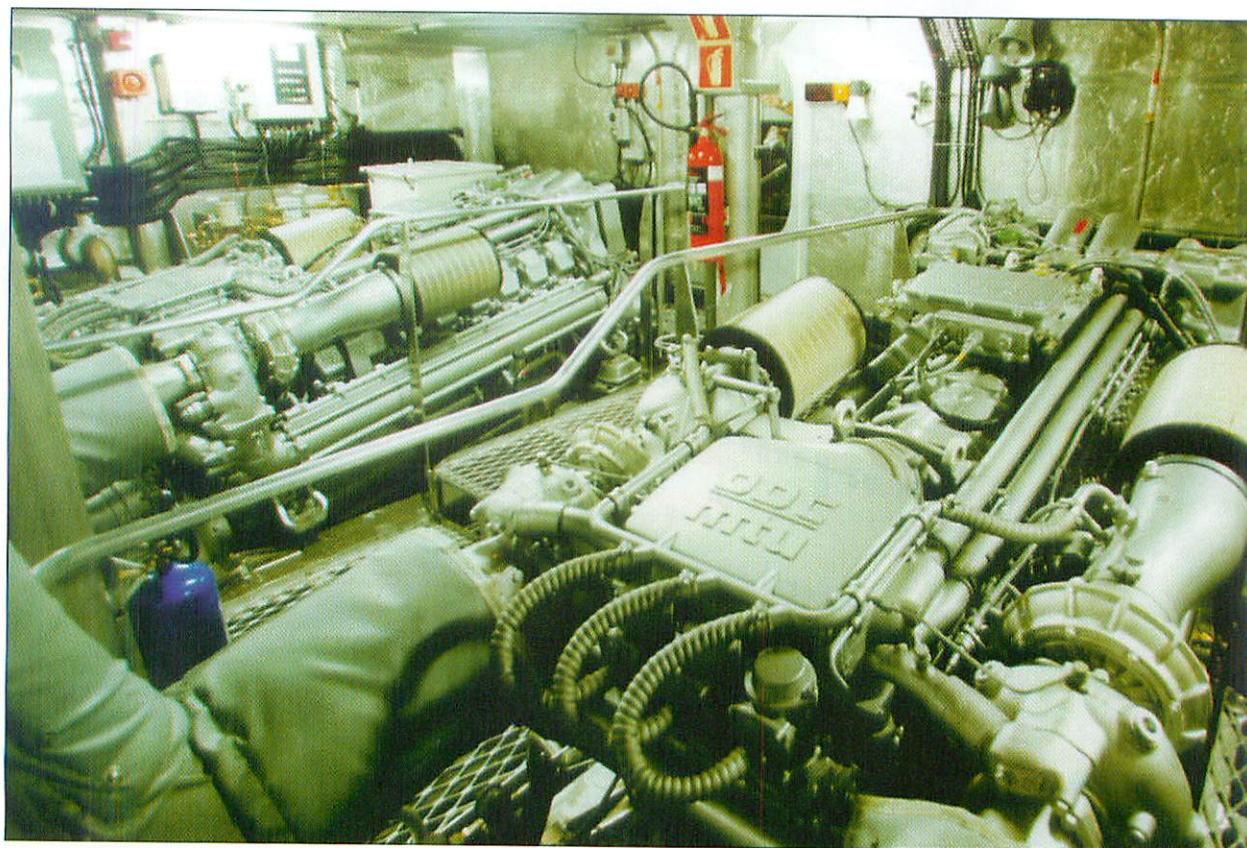
A major feature of the Bay Class vessels is the tender launch and recovery system. Two 6.4m centre console RIBs with a deep-V hullform are kept in Vest davits on the aft deck. A single point release system, capable of being safely connected and disconnected from the davit cable by a single crewman in all conditions allows the tenders to be launched or recovered at speeds up to five knots. Powered by twin 90hp Honda outboards, the RIBs can be deployed, fully loaded in up to Sea State 4.

The central elevated bridge provides 360 degree vision and is outfitted with an

engineering station, helm position, officer of the watch position and master's station. Main propulsion and bow thruster controls are repeated at the wing stations, where a clear view of the tender launch and recovery areas can be obtained. All machinery and monitoring systems on board the vessel are controlled from the engineer's station. The bridge is equipped with two Leica DGPS units, a Transas ECDIS display, Racal Decca 3cm and 10cm radar and ARPA and a C Plath gyro/autopilot. Aft of the bridge is an ergonomically designed communications station housing a GMDSS system for area A3, CYCOM field data terminal and printer, separate UHF for communicating with both Customs and the Australian Defence communications system and an encrypted satellite fax.

Accommodation, which includes six lockable two-berth cabins, operations room, office, interview room, galley and mess, is both comfortable and easy to maintain. Noise levels do not exceed 70dBA in any internal areas and are limited to 65dBA in the bridge and interview/recreation room. The latter is located amidships on the main deck and is capable of simple conversion to sleep four people. It is equipped with seating for eight, interview recording equipment and an audio visual system.

The large and fully equipped galley is



located forward on the main deck. The mess consists of two tables and seating for twelve and also houses a secure armoury for rifle and side arm storage. Adjacent to the galley is a large freezer, refrigerator and dry storage facility, capable of carrying sufficient stores for 12 people for 28 days. Dimmable red lighting is fitted throughout the vessel and the craft can be totally blacked out if necessary.

Deck layout is designed to comply with occupational health and safety principles and to facilitate all possible combinations of vessel functions. The aft main deck area is well equipped and ease of crew movement in this area is assisted by well placed handrails and bulwarks. Railings and gateways are arranged to facilitate the recovery of objects from the water, removal/replacement of equipment from

the engine room and loading/unloading of cargo. Recessed tie downs are fitted on the main deck to enable secure cargo handling. Cargo and vessel machinery can be handled via a marine cargo crane situated on the upper deck, aft of the operation's room.

Vessel towing is facilitated by three towing bollards across the transom. The centre aft bollard has a towing capacity of 12 tonnes. A further ten bollards are situated around the main deck to enable easy docking at any type of wharf in all weather conditions. Mounted aft of the transom is a full width dive platform with access via permanent stairs and a transom gate to support SCUBA diving operations.

Aft of the superstructure is a permanent semi-enclosed area with removable bench seating for 12 additional passengers.



Main Particulars

Owner

Australian Customs

Builder

Austal Ships, Australia

Length oa 38.2m

Length wl 32.1m

Beam mld 7.20m

Draft (hull) 2.2m

Main engine

2 x MTU 16V 2000 M70 diesels,
1,050kW each

Propulsion

2 x Veem four-bladed propellers
driving through Reintjes
WVS430/1 gearboxes

Speed 21 knots

Range 1000nm

Crew 4

Accommodation 12

Classification

DNV +1A1 LC R1 Patrol EO

Equipment

Vest davits, Leica DGPS, Transas
ECDIS, Racal Decca radar and
ARPA, C Plath gyro/autopilot

*Above: A pair of MTU diesels
dominate the engine room.*

*Left: The central elevated bridge
has all round vision.*



STILLWATER RIVER: SWATH design for smooth crew transfer

The all-aluminium 36.58m x 13.10m SWATH crewboat *Stillwater River* has more than fulfilled operating requirements set down by Brazilian offshore company Petrobras. Built in Florida at Eastern Shipbuilding for Trico Marine Services of Louisiana and put into service with Petrobras transporting crew to rigs in the Campos Basin, the vessel design was subjected to intensive testing to prove its seakeeping qualities.

The initial specification from Petrobras, presented to Houston based naval architects Alan C McClure Associates (ACMA), is worthy of report. It stated that the vessel must be able to maintain an average speed in excess of 25 knots with less than five per cent incidence of motion sickness during four-hour voyages in 3m seas and 25-knot winds. ACMA, whose founder, the late Alan McClure, was a pioneer of SWATH design in the early sixties, did not hesitate to recommend this hull form as offering the best chance of meeting the criteria. Model testing, observed by Petrobras, confirmed the rationale. Waves had a minimal effect on the vessel's amplitude and frequency of motion and forward speed.

Characteristically, 80 per cent of the vessel's displacement is provided by twin cylindrical submerged hulls, which support the main hull and superstructure on slender

columns. These lower hulls each house a 4,600kW Allison 501-KF5 marinised gas turbine coupled to a Philadelphia 20MGH-HP2S gear with a 28:1 reduction ratio turning 1,828mm diameter four-bladed Lips CP propellers through conventional shafting. According to ACMA, installing the turbine/gear arrangements in the submerged hulls, despite the restricted space, considerably improves stability, more than counteracting the effect of the relatively large windage area.

At the bow of the port hull there is a Holland Roer Propeller tunnel thruster with 812mm propeller driven by a 200hp variable speed electric motor, to assist station-keeping when unloading and loading rig crews.

Four custom-made active non-retractable fins are fitted on the inboard side of both lower hulls at bow and stern and controlled by a Maritime Dynamics computer-based system, which monitors acceleration, through sensors located at key points around the vessel.

Aircraft-style seating is provided on two decks with a 3-4-4-3 arrangement on the main deck to accommodate 196 with a further 54 seats in a 3-3-3-3 layout, divided by the stairs, at the forward end of the upper deck. The aft end is taken up by an offshore transfer zone flanked each side by the turbine exhausts and aft-facing

Peerless engine air intakes with dewatering filters. Dockside boarding and disembarkation is via the main deck.

Quarters for six crew (captain, mate, two engineers and two deck hands) are provided in four cabins situated forward of the passenger area under the pilothouse. Passenger toilet spaces are also grouped forward, port and starboard, and a galley, positioned amidships on the main deck, dispenses airline-type meals. Accommodation areas are air-conditioned throughout.

The pilothouse has forward and aft helm positions with a joystick in the armrest of both chairs. These can control the rudders, throttles, pitch of the CPP and operation of the bow thruster through another microprocessor system. A full GMDSS communication set-up including Inmarsat C, featuring components mainly provided by SEA, is installed. For navigation, two ARPA interfaced Simrad Anritsu radars, Trimble GPS, Sperry gyro and Simrad Taiyo direction finders are fitted.

Although operating off Brazil, the vessel is US-flagged, classed by ABS and satisfies both USCG Small Passenger Boat and MARPOL regulations. ACMA provided construction, dock and sea trial supervision and all the engineering required to obtain the regulatory and classification approvals.

STILLWATER RIVER

Main Particulars

Owner

Trico Marine Services, USA

Builder

Eastern Shipbuilding, USA

Designer

ACMA, USA

Operator

Petrobras, Brazil

Length 36.58m

Beam 13.10m

Passengers 250

Speed 29 knots max

Main power

2 x Allison 501-KF5 gas turbines,
4,600kW each

Auxiliary power

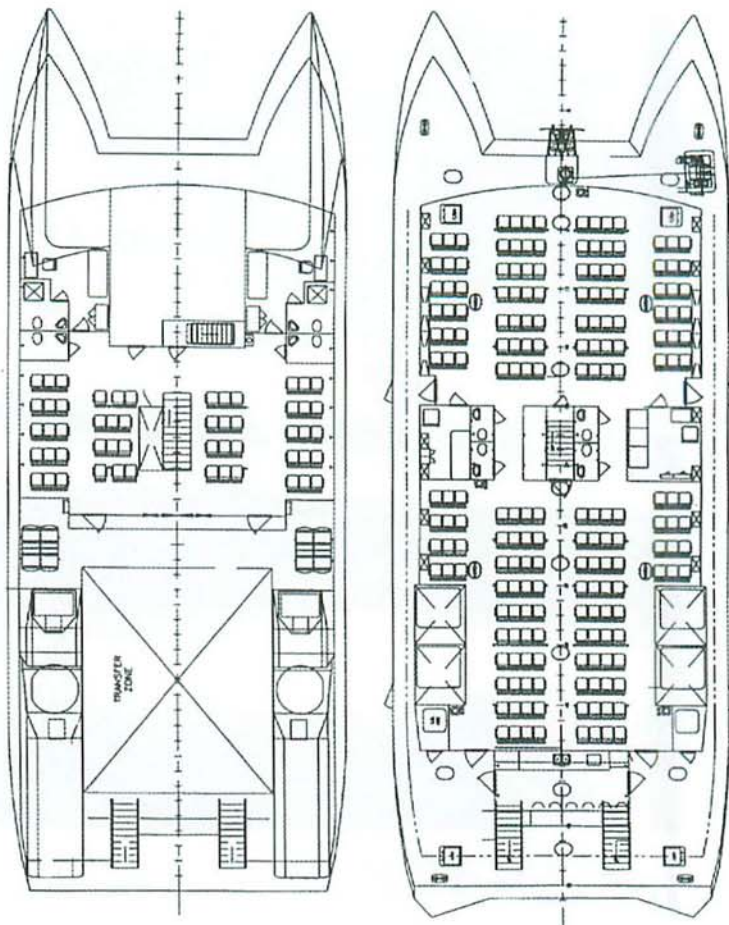
2 x 315kW Cummins NTA 855
driven gensets

Propulsion

4-bladed Lips CP propellers
through conventional shafting,
Holland Roer Propeller tunnel
thruster with 812mm propeller

Equipment

SEA GMDSS system, Simrad
Anritsu radars, Trimble GPS,
Sperry gyro, Simrad Taiyo
direction finder



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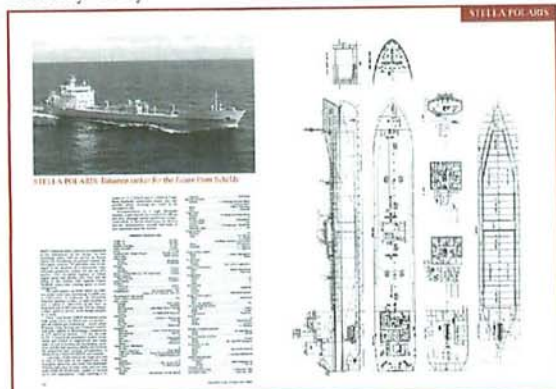


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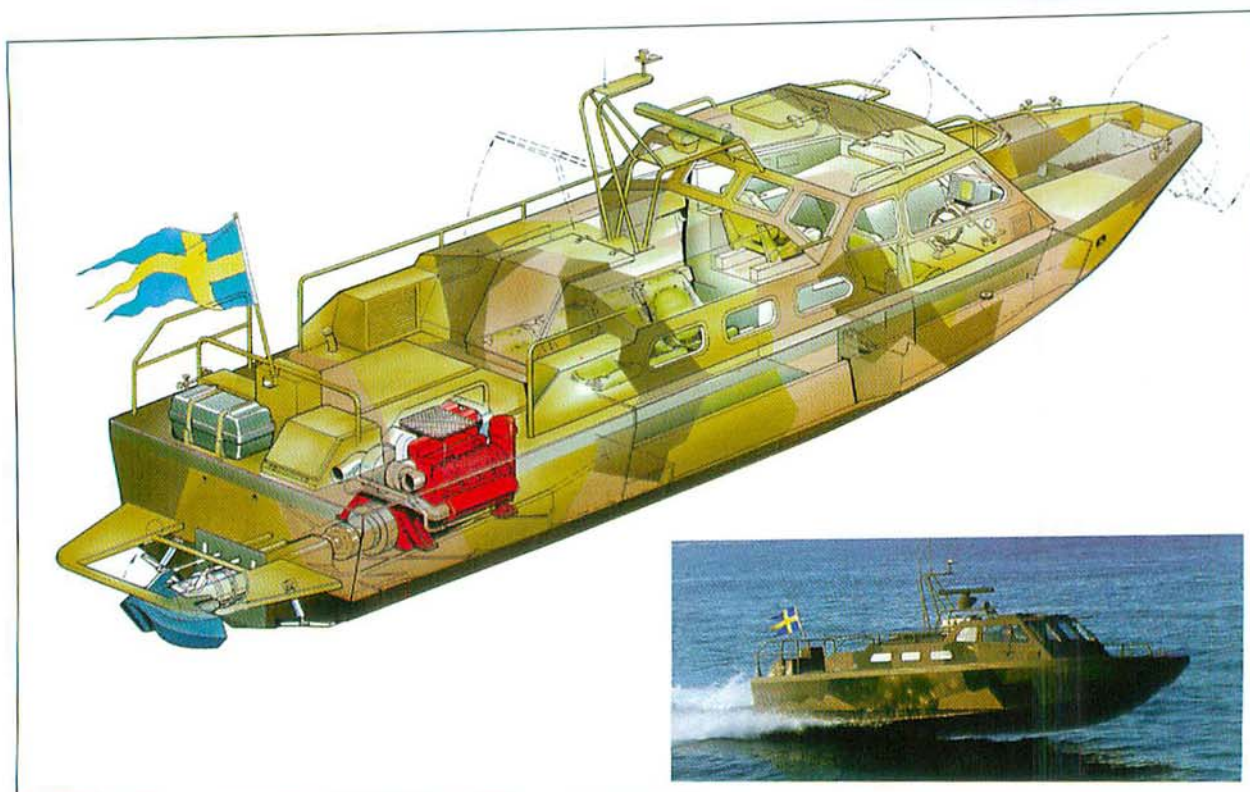
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STOREBRO 90E: Swedish success in commercial market

A fast combat boat from Swedish builder Storebro in use with the Swedish Navy, makes for a surprisingly versatile workboat if deprived of its military livery. The craft, designated 90E, is being marketed by the yard's commercial division, which is well known for its range of luxury motor boats.

Measuring 11.88m x 2.90m, the 45-knot vessel is of composite construction and is said to accept 4g stresses whilst isolating occupants from shock in a sound-proofed interially suspended cabin.

Two civilian versions were built in '99 for very different roles. Oslo Municipality in Norway needed a combined rescue, ambulance and fire-fighting boat, while the joint US-Lithuanian company Mazeikiu Nafta required a vessel capable of serving offshore in demanding Baltic conditions.

Essentially, both craft are unchanged from the standard navy model with the same low-weight hull and diesel/waterjet combination providing sparkling manoeuvrability and performance. However, they are customised very differently. Oslo's boat, which was delivered in July '99, is fitted with the same standard and level of medical treatment equipment as an advanced road ambulance with an identical stretcher system. In addition, it has a powerful 1,000 litre/min water pump for fire-fighting and pumping out other vessels and, for the search and rescue role, a 360-degree remotely controlled searchlight mounted on a 5m

telescopic mast. An aft platform provides a rescue station and gives protection to the jet bucket.

The second craft, delivered during the same month, is now at work 10km off the Lithuanian coast at Klaipeda, helping to load and unload mooring buoys. An added continuous fender system both sides of the hull permits mooring alongside tankers even in a considerable swell. Special requirements of Mazeika Nafta Oil included a two-tonne bollard pull, a net system to lash down buoys and other loads on the open deck aft and an AC power supply able to operate onboard computers.

Of monohedral V-form, the hull provides good seakeeping qualities in choppy Baltic seas. Nevertheless, a modest draft of only 0.7m, along with waterjet propulsion, permits operation in shallow waters near islands. Keel and chines are reinforced for high-speed beach landings. A crew of two can operate the vessel, which is capable of accepting loads of up to two tonnes, five stretcher cases or ten seated personnel.

Hans Lindgren, general manager of the commercial vessel division at Storebro, is well pleased with the success to date for the 90E, believing that these deliveries have opened the door to a future in the non-military sector. A standard Swedish Navy Craft displayed in the UK at the Seawork exhibition, demonstrated an impressive manoeuvrability with tight high-speed turns, rapid acceleration and

the ability to crash stop from high speed within a single boat length. Maximum speed with a single Scania DSI 14 V8 diesel, coupled to a Kamewa FF 410 waterjet, is some 45 knots. Range at over 40 knots is 215nm.

Main Particulars

Builder

AB Storebro Royal Cruiser, Sweden

Designer

Swedish Defence Material Administration in co-operation with Swedish Coast Artillery & Storebro

Length oa 11.88m (with jet)

Length hull 10.80m

Beam 2.90m

Draft 0.70m

Main engine

Scania DSI 14 V8 diesel, 460kW

Propulsion

Kamewa FF 410 waterjet

Speed 40 knots plus

Crew 2

Passengers 10 max.